3-D Receiver Operating Characteristic Analysis for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image classification (HSIC) faces three major challenging issues, which are generally overlooked. One is how to address the background (BKG) issue due to its unknown complexity. Another is how to deal with imbalanced classes since various classes have different levels of significance, particularly, small classes. A third one is fractional class membership assignment (FCMA) resulting from a soft-decision classifier. Unfortunately, the commonly used classification measures, overall accuracy (OA), average accuracy (AA), or kappa coefficient are generally not designed to cope with these issues. This article develops a 3-D receiver operating characteristic (3-D ROC) analysis from a detection point of view to explore how these three issues can be resolved for HSIC. Specifically, it first develops one-class classifier in BKG (OCCB), called constrained energy minimization (CEM), and multiclass classifier in BKG (MCCB), called linearly constrained minimum variance (LCMV) in conjunction with 3-D ROC analysis to address the BKG issue. Then, by considering a small class as a signal to be detected, its class accuracy can be interpreted as signal detection power/probability so that the 3-D ROC analysis can be used to address the imbalanced class issue. Finally, FCMA can be treated as a detector by converting a soft-decision classifier to a hard-decision classifier in such a manner that the 3-D ROC analysis is also readily applied. The experimental results demonstrate that 3-D ROC analysis provides a very useful evaluation tool to analyze the classification performance.**

***Index Terms*— 3-D receiver operating characteristic (3-D ROC), fractional class membership assignment (FCMA), hyperspectral image classification (HSIC), iterative constrained energy minimization (ICEM), iterative edge preserving filter (IEPF), iterative linearly constrained minimum variance (ILCMV).**

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Nomenclature

3-D ROC	3-D receiver operating characteristic.
AA	Average accuracy.
AUC	Area under curve.
AVIRIS	Airborne Visible/Infrared Imaging Spectrometer.
BKG	Background.
CEM	Constrained energy minimization.
EPF	Edge preserving filter.
FCMA	Fractional class membership assignment.
HSIC	Hyperspectral image classification.
ICEM	Iterative CEM.
IEPF	Iterative EPF.
ILCMV	Iterative linearly constrained minimum variance.
LCMV	Linearly constrained minimum variance.
MAP	Maximum *a posteriori*.
MCCB	Multiclass classifier in BKG.
OA	Overall accuracy.
OCC	One-class classification.
OCCB	One-class classifier in BKG.
OPR	Overall PR.
PR	Precision.

I. Introduction

HSIC has received considerable interest in recent years, and many efforts have been reported in the literature, such as support vector machine (SVM)-based HSIC [1]–[3], neural network-based classification [4], kernel-based HSIC [5], [6], sparse representation-based HSIC [7]–[9], deep learning network-based HSIC [10]–[15], an efficient and effective semisupervised learning approximation algorithm of hypergraph p-Laplacian to preserve the geometry of the probability distribution [16], a hyperspectral image unsupervised classification framework based on robust manifold matrix factorization and its out-of-sample extension [17], finding simultaneous spectral–spatial (SS) feature selection, and extraction algorithm for hyperspectral image SS feature representation and classification [18]. Of particular interest are SS-based HSIC techniques [19]–[29], which include a spectral classifier in conjunction with a spatial filter to extract pixel spatial information to improve classification. For example, Tarabalka *et al.* [2] used segmentation to obtain spatial information by Markov random field (MRF) regularization, and Zhong *et al.* [12] took care of spectral and spatial

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information simultaneously by using a 3-D residual network. Specifically, we are interested in the approach developed in [23], called EPF-based methods, which used SVM to perform spectral classification followed by spatial domain-based techniques to capture spatial contextual information from SVM-classification maps. Such spatial information obtained directly from SVM classified data samples is referred to as *a posteriori* spatial information.

However, three major issues have been overlooked in HSIC and not received much attention in the past. One is how to factor the BKG issue into HSIC. Over the past years, the BKG issue has not been a major concern in HSIC. This is because HSIC has been performed with the provided ground truth and the BKG is knowingly excluded from classification due to the fact that BKG contains so many unknown signatures, which make BKG very complicated for data analysis. Hence, if BKG is considered part of classification, the commonly used classification measures, OA, AA, or kappa coefficient may not be effective since they do not take into account the effects caused by BKG.

A second issue is how to address the issue of imbalanced classes, which was also investigated in [27]. In other words, different classes may have various sizes of data samples, some of which may be very large, while some of which may be very small. Consequently, the classification accuracy produced by large classes may dominate and overwhelm the classification accuracy of small classes if an appropriate measure is not taken. Hence, a larger class does not necessarily imply that it is more significant than a smaller class. As a matter of fact, in many real-world applications, small classes are very likely to be more important than large classes. For example, rare material substances are generally most vital classes in hyperspectral data but their populations are usually very small. So, when they are present in the data, the samples of these substances are not very large. Accordingly, a small class does not imply its importance less than a larger class. The commonly used sample ratio (SR) that calculates the ratio of the number of samples in one class to the total number of data samples may not be appropriate to measure class significance since it places more weights on larger classes than smaller classes. Therefore, how to address unbalanced classes in different sizes poses a great challenge in classification.

A third issue is how to deal with FCMA resulting from a classifier, which makes soft decisions. Because of acquisition from hundreds of contiguous spectral bands, one of the major strengths provided by hyperspectral imaging sensors is a wealth of spectral information containing abundance fractions of subtle materials that generally cannot be captured by multispectral imaging sensors. Unfortunately, classification is a hard-decision-based class membership assignment. Hence, how to convert soft-decision-based classification to a hard-decision-based classification is challenging when an FCMA is generally needed for HSIC. This is indeed a main issue arising in HSIC. To mitigate such problem, one common approach is SS-based methods [19]–[29], which use a spectral method such as SVM to perform a hard-decision spectral classification. However, the spectral classifier used by this SS approach apparently performs pure pixel classification to produce hard-decision maps. Since such a pure pixel-based spectral classifier does not explore spectral abundance fractions of material substances present in a pixel to resolve this issue, spatial filters are then used as a follow-up process to capture spatial information of spectral-classified data samples to generate soft-decision classification maps. This gives rise to an issue of how to convert spatial-filtered soft-decision classification maps to hard-decision classification maps. For example, if there is one data sample in the soft-decision classification maps that contains 45% of one substance, 40% of another one, and 15% for rest of substances, a natural hard decision may be forced by the majority vote such as MAP [30], which assigns the data sample to the class of the substance with 45%. However, it may lead to an incorrect hard decision. This scenario also occurs in multiclass SVM classification, where one-against-one rule or one-against-rest is used for classification of multiple classes [30].

Interestingly, the abovementioned issues have not received much attention in remote sensing community. For the BKG issue, an approach, called OCC, was developed to perform in two ways [31]–[42]. One is binary classification by treating a class of interest as one single class and all other classes as another class. The other is to only consider the classification of a single class of interest, while discarding other classes. There is a difference between these two approaches, which is the use of training data where binary classifiers require an exhaustively labeled training data set while one-class classifiers are trained only using samples of the class of interest. Most recently, Hossain *et al*. [41] made use of the cluster representation space to model BKG to perform OCC. Unfortunately, the BKG referred in OCC is generally made up of uninteresting classes but not including real BKG of image data, for example, class 17 in the Purdue data [see Fig. 10(b) in Section VIII-A]. Hence, technically speaking, OCC still cannot resolve the BKG issue.

This article takes a rather different route from the OCC approach. Its idea is derived from statistical detection theory. It considers an OCC problem as a one-class detection (OCD) problem. The key difference between OCC and OCD lies the fact that OCC performs as a classifier that requires either modeling BKG by training samples or probability distributions compared to OCD which performs as a detector without knowing BKG. To carry out OCD, a recently developed SS approach, called ICEM in [40], can be used as a desired detector, where the class of interest is considered as the desired signal to be detected in BKG specified by the alternative hypothesis H_1 and all other classes plus BKG as the null hypothesis H_0. Accordingly, the BKG issue can then be taken care of by ICEM. By singling out a class of interest as the desired signal to be detected, the issue of classification of imbalanced classes can be resolved. This is because all other classes including large classes are considered as BKG. In this case, ICEM can be regarded as a counterpart of OCC. ICEM was later extended to multiple-class detection, called ILCMV, which has no counterpart in OCC.

Once one-class or multiple-class detection is performed by ICEM or ILCMV, an evaluation criterion is needed to measure its detection performance for one class or multiple

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classes. This requires a threshold τ for a detector to calculate detection probability and false alarm probability. Apparently, the commonly used OA and AA for OCC are not applicable to evaluating a detector since the detection performance is determined by the threshold τ and the traditional ROC analysis can be used as an evaluation tool. However, classification is a hard-decision maker and can only produce a single point of (P_D, P_F) on an ROC curve. This may be the reason why ROC analysis has not been considered as a measure for classification.

One significant drawback of using ROC analysis is its ROC curve, which only describes the performance of P_D relative to P_F and does not address P_D and P_F separately. As a matter of fact, both P_D and P_F are indeed dependent functions of the threshold τ as an independent parameter that is determined by a given upper bound β on false alarm probability, P_F. If we interpret P_D as a function of the parameter τ, then P_D can be used to measure class accuracy, P_A, in classification.

Recently, a 3-D ROC analysis was particularly developed to resolve the abovementioned issue [43]–[45] by expanding the traditional 2-D ROC curve of (P_D, P_F) to a 3-D ROC curve of (P_D, P_F, τ), where the threshold τ is now considered as an independent parameter and both P_D and P_F can be considered as functions of τ. By virtue of a 3-D ROC curve, three 2-D ROC curves, 2-D ROC curve of (P_D, P_F), 2-D ROC curve of (P_D, τ), and 2-D ROC curve of (P_F, τ) can be generated. Specifically, the 2-D ROC curve of (P_D, τ) can be used to evaluate P_A of a classifier and 2-D ROC curve of (P_F, τ) can be used to assess the effect of BKG on classification. Hence, varying values of τ allows us to generate a 3-D ROC curve. Specifically, for each pair of (P_D, P_F), it corresponds to one single point in 2-D ROC curve specified by the same particular threshold value τ.

This article utilizes 3-D ROC analysis as an evaluation tool to effectively address the abovementioned three issues caused by BKG, imbalanced class, and FCMA for HSIC. To accomplish this goal, a binary classification problem is first translated to a binary hypothesis testing problem from a signal detection (SD) point of view in Section III, where a two-class confusion matrix can be interpreted by P_D and P_F. Then, a multiclass classification problem is further extended to a multihypotheses testing problem in Section IV, where a multi-class confusion matrix can also be derived from the multitarget detection probability and false alarm probability. In particular, the threshold τ can be used to address the FCMA issue.

As a final note, this article can be considered as a companion article of a recent work [46], which solely focused on using statistical detection theory approach to evaluate HSIC via P_{OA} and P_{PR} but did not use 3-D ROC analysis as performance measure. This article used in conjunction with [46] should provide a clear picture of how classification can be treated and analyzed by the well-established statistical detection theory.

II. Novelties and Contributions

One of the major novelties/contributions of this article is to reinterpret the traditional detection-based ROC analysis as an evaluation tool for classification performance from which three other novelties/contributions can also be derived to address three issues arising in HSIC, each of which is described in detail as follows.

1) *The 3-D ROC Analysis for Classification Performance*: A key idea of taking advantage of ROC analysis is to interpret a class to be classified as a signal while other classes plus BKG as noise and formulates an OCC problem as a binary composite hypothesis testing problem to convert a classifier to a detector. Specifically, by taking advantage of the Neyman–Pearson detection theory, the false alarm probability, β, can be used to determine a threshold τ, which is in turn to calculate P_D and P_F. Using the β-determined threshold τ as an independent parameter along with P_D and P_F as functions of τ, we can generate a 3-D ROC curve of (P_D, P_F, τ) from which three 2-D ROC curves, 2-D ROC curve of (P_D, P_F), 2-D ROC curve of (P_D, τ), and 2-D ROC curve of (P_F, τ) can also be generated. As a result, the AUC of 2-D ROC curve of (P_D, τ) and the AUC of 2-D ROC curve of (P_F, τ) can be interpreted as class accuracy and misclassification (MC) error of a classifier, respectively, while the AUC of the 2-D ROC curve of (P_D, P_F) can be used to evaluate the effectiveness of a classifier. Specifically, it should be noted that unlike the traditional ROC curve of (P_D, P_F) that are actually calculated by the same threshold τ predetermined by a prescribed upper bound β on false alarm probability, a 3-D ROC curve is in fact a plot of three parameters (P_D, P_F, τ), where the 2-D ROC curve of (P_D, τ) and the 2-D ROC curve of (P_F, τ) are actually calculated by τ independently. The concept of using AUC values of these two curves to measure class accuracy and MC error is new and has never been explored in the literature. It is also known that the traditional 2-D ROC curve of (P_D, P_F) is generated by varying P_F as a variable, where there is no clue about what the threshold τ is to produce P_F. By contrast, the 2-D ROC curve of (P_D, τ) and 2-D ROC curve of (P_F, τ) vary with their own individual threshold values of τ and both are independent of each other. A 3-D ROC curve provides individual 2-D ROC curve profiles of P_D and P_F versus different values of τ, which do not exist in the traditional 2-D ROC analysis. This may explain why the 2-D ROC analysis has received little attention in classification.
2) *BKG Issue*: Since HSIC is generally performed with the provided ground truth, BKG is intentionally excluded from classification. This is because the complexity of BKG is very difficult to be characterized. If BKG is included in classification, an immediate arising issue is how to evaluate its performance. After all, there are so many unknown signal sources present in BKG without prior knowledge that can have significant impact on classification. Accordingly, many techniques developed for HSIC have not considered BKG as a factor in HSIC. By realizing the role that BKG plays in HSIC, recent research works proposed OCC approaches, which only classified a class of interest while either considering all other classes as another class to perform two-class

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classification or simply discarding all other classes without classification. One novelty of this article models such an OCC problem as an SD problem, where the class of interest is considered as an alternative hypothesis representing a signal to be detected and all other classes including BKG as a null hypothesis corresponding to noise. So, with this binary hypothesis testing formulation, the problem of classifying a class of interest can be translated to a problem of detecting a signal in noise so that the well-established ROC analysis can be readily applied to OCC. In particular, the P_F can be interpreted as the probability of misclassifying BKG misclassified into the class of interest as the signal. In this case, the 2-D ROC curve of (P_F, τ) can be used to address the impact of BKG suppression on classification, which has never been investigated in the past.

3) *Imbalanced Class Issue*: In HSIC, two commonly used classification measures are OA and AA, both of which do not work effectively for small classes due to the fact that OA is calculated based on SR (i.e., the ratio of data samples in a class to the entire data samples) and AA averages the accuracies of all classes. So, when a class contains a very small number of data samples, its classification accuracy can be very easily sacrificed or compromised by the accuracy of a large class as shown in [47] and [48]. Although such an imbalanced issue was addressed by the concept of class significance calculated by class features proposed in [48], a second novelty of this article investigates this issue from a Neyman–Pearson detection theory aspect. If we consider a small class as a signal to be detected while all other classes plus BKG as noise, then its class accuracy can be interpreted as SD power. In this case, the detectability of a small class can be calculated by a specific detection probability, P_D. The difference in detection between a small class and a large class is that a small class is considered as a weak signal as opposed to a large class which can be considered as a strong signal. In this case, the 2-D ROC curve of (P_D, τ) can take care of detection of a small class to resolve the imbalanced class issue.
4) *FCMA Issue*: Due to the use of hundreds of spectral bands provided by a hyperspectral imaging sensor, the abundant spectral information can be very useful in classification. Unfortunately, classification is generally performed by hard decisions and a direct use of hard-decision classifier for classification is usually not effective. In order to resolve this situation, a spatial filter is generally incorporated to capture spatial information which results in real values that lead to soft decisions. So, how to evaluate a soft-decision classifier becomes very challenging. Hence, a third novelty of this article converts a soft decision to hard decision by varying a threshold via 3-D ROC curves. In this case, the 2-D ROC curve of (P_D, P_F) can be used to address the effectiveness of a classifier.
5) As a summary, the contributions of this paper are the abovementioned novelties to utilize the three 2-D ROC curves of (P_F, τ), (P_D, τ), and (P_D, P_F) to resolve the BKG issue, imbalance class issue, and FCMA issue, respectively, each of which is generated from a 3-D ROC curve. Most importantly, the 3-D ROC analysis is developed as an evaluation tool to measure classification performance regardless of how a classifier is designed. Accordingly, it can be applicable to any soft-decision classifier, particularly for SS classifiers, such as EPF [23], IEPF [49], iterative SVM [50], ICEM [28], and ILCMV [29] as demonstrated in the experiments conducted in this article. Nevertheless, we should point it out that the 3-D ROC analysis cannot be directly applied to convolutional neural network (CNN)-based classifiers [10]–[15] due to its hard decisions made at the end of fully connected neural networks. To make the 3-D ROC analysis work for CNN, we need to look into the soft decisions made by CNN before it makes hard decisions at the end. Since this issue is beyond the scope of this article, it will be a new direction to be investigated in the future.

III. Binary Hypothesis Testing

Generally, a binary classification can be formulated as a binary hypothesis testing problem. It is natural to take the well-developed detection theory to address the abovementioned issues for classification problems. A recent work applying statistical detection theory to multiclass hyperspectral classification was investigated in [46].

An SD problem is formulated by a binary hypothesis testing problem as a binary classification problem.

A. Neyman–Pearson Detection Theory

An SD problem is generally described by a binary hypothesis testing problem [51] as follows:

$$\begin{aligned} &H_0 : Y \approx p_0(y) \\ &\text{versus} \\ &H_1 : Y \approx p_1(y) \end{aligned} \tag{1}$$

where $p_0(y)$ and $p_1(y)$ are the probability density functions (pdfs) that govern the two hypotheses H_0 and H_1, called "null hypothesis" that represents the signal absence and "alternative hypothesis" that represents the signal presence, respectively. A Neyman–Pearson detector (NPD), denoted by $\delta^{NP}(\mathbf{r})$, is designed to solve the following constrained optimization problem:

$$\begin{aligned} &\max_{\delta} \ \{P_D(\delta)\} \\ &\text{s.t. } P_F(\delta) \le \beta \text{ for any given } 0 \le \beta \le 1 \end{aligned} \tag{2}$$

where β is prescribed as the significant level of test. The maximum in (2) is sought over all possible detectors, $\delta(\mathbf{r})$. The well-known Neyman–Pearson lemma shows that the optimum solution to (2) is a likelihood ratio test, $\Lambda(\mathbf{r})$, given by

$$\delta^{NP}(\mathbf{r}) = \begin{cases} 1; & \text{if } \Lambda(\mathbf{r}) > \tau \\ 1 & \text{with probability } \gamma \text{ if } \Lambda(\mathbf{r}) = \tau \\ 0; & \text{if } \Lambda(\mathbf{r}) < \tau \end{cases} \tag{3}$$

where $\Lambda(\mathbf{r}) = p_0(\mathbf{r})/p_1(\mathbf{r})$, the threshold τ is determined by the constraint β, and γ is the probability of saying H_1 when

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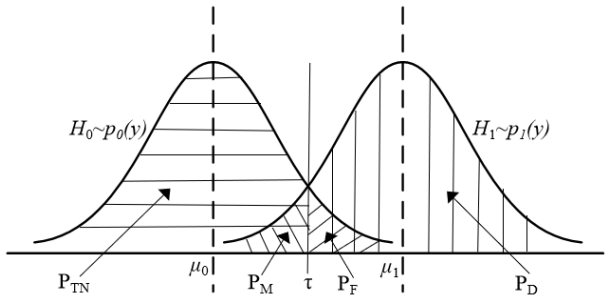


Fig. 1. Illustration of probabilities P_D, P_F, P_M, and P_{TN}.

TABLE I
CONFUSION MATRIX FOR SIGNAL-NOISE TWO-CLASS CLASSIFICATION

classification \ true	noise	signal
noise	P_{TN}	P_M
signal	P_F	P_D

$\Lambda(\mathbf{r}) = \tau$ given by $\gamma = (\beta - P_0(\{y \in \Gamma | p_1(y) > \tau p_0(y)\}))/(P_0(\{y \in \Gamma | p_1(y) = \tau p_0(y)\}))$. Using (3), the P_D and P_F can be found by

$$P_D = \int_{\Lambda(\mathbf{r})>\tau} p_1(\mathbf{r})d\mathbf{r} + \gamma\, P(\{\mathbf{r}|\Lambda(\mathbf{r}) = \tau\}) \quad (4)$$

$$P_F = \int_{\Lambda(\mathbf{r})>\tau} p_0(\mathbf{r})d\mathbf{r} + \gamma\, P(\{\mathbf{r}|\Lambda(\mathbf{r}) = \tau\}) \quad (5)$$

with the τ determined by $P_F = \beta$ via (5). As an illustrative example, Fig. 1 shows four decisions made by NPD, where $P_{TN} = 1 - P_F$, referred to as true negative (TN) corresponding to the signal absence and $P_M = 1 - P_D$ referred to as miss probability corresponding to no SD when there is a signal present. Details of the Neyman–Pearson detection theory can be found in [51].

According to Fig. 1, we can construct the following matrix in Table I, which corresponds to a confusion matrix commonly used in binary classification.

B. ROC Analysis

According to (2), the detection performance of an NPD in (3) is determined by P_D subject to the constraint on P_F bounded from above by a specific given value β. To be precise, for a given β, we can calculate P_F using (5) to find the threshold τ. This same τ is further used to calculate P_D via (4). So, a different value of β also produces different values of P_F and P_D. If we now vary the value of β from 0 to 1, then we can plot a curve of P_D versus P_F, as shown in Fig. 2, which is called ROC curve.

This ROC curve allows users to assess the effectiveness of a detector as a function of P_D versus P_F. However, on some occasions, two detectors δ^1 and δ^2 may generate two different 2-D ROC curves but have the same area under their ROC curves, A_z, as shown in Fig. 3.

In this case, both δ^1 and δ^2 may yield the same detection performance even they have different ROC curves. To address this dilemma, an alternative approach to the use of ROC curves is to calculate the area under an ROC curve, called under curve (AUC), A_z instead, which is particularly useful in medical diagnosis [52]. This AUC can be used to evaluate

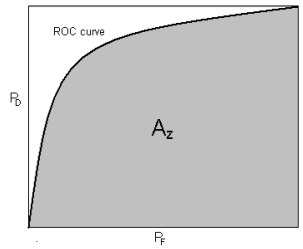


Fig. 2. Example of AUC, A_z, calculated by an ROC curve.

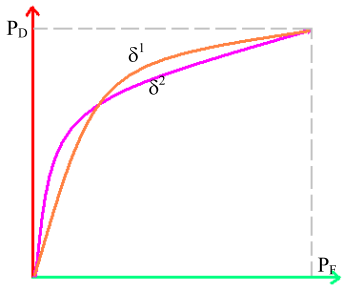


Fig. 3. Example of ROC curves generated by two detectors with the same A_z.

the effectiveness of a detector. In other words, using AUC as a criterion, the two detectors δ^1 and δ^2 in Fig. 3 indeed have the same effectiveness of detectability by producing the same A_z even though they generate two different ROC curves. Also, a comment on an ROC curve is noteworthy. For each value of β, it generates a pair of (P_D, P_F) as a point at an ROC curve, where a threshold τ can be found by (5) by setting $P_F = \beta$ and then both P_D and P_F are calculated by the same τ produced by (4) and (5).

One major advantage of using ROC analysis is no requirement of prior knowledge about the cost and probabilities of hypotheses, H_0 and H_1. This is exactly why ROC analysis is a perfect candidate that can be used to evaluate the impact of BKG in classification as will be discussed in Section III-C.

C. Applications to Signal Classification in BKG

Now, if we replace the hypotheses H_0 and H_1 in (1) with two classes, signal class and no signal class, respectively, where no signal class can be considered as BKG to represent "all other classes plus BKG" in H_0, then we can re-express the hypothesis testing problem in (1) as detection of desired signal class in BKG by

$$H_0 : \text{BKG} \approx p_0(y)$$
versus
$$H_1 : \text{signal} + \text{BKG} \approx p_1(y). \quad (6)$$

A major strength of a binary hypothesis testing problem is to detect signals in noise. By virtue of detection theory, we can interpret signals as a class of interest and all other signal sources along with noise as BKG. As a consequence, the detection theory can be used to classify imbalanced classes since each individual class will be considered as a signal of interest specified by H_1 and all other classes plus BKG will be lumped into the BKG class specified by H_0. Especially, a small class can be considered as a signal of interest to be

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TABLE II
CONFUSION MATRIX FOR SIGNAL-BKG TWO-CLASS CLASSIFICATION CLASSIFICATION

classification \ true	signal	BKG (no signal)
signal	SD (signal detection)	FS (false alarm)
BKG (no signal)	FB (miss signal)	BD (no signal)

detected in BKG. In this case, it will not be compromised by large classes, which are included as part of BKG.

Now, we can consider a class of interest and BKG as two separate classes specified by two hypotheses, alterative hypothesis H_1 and null hypothesis H_0, respectively. Clearly, these two classes are imbalanced with BKG being a large class and the signal being a small class. Most importantly, such a small signal class is crucial in SD. Hence, signal classification can be performed like SD by considering BKG as a class of no interest. In this case, only detection probability $P_{\rm D}$, false alarm probability $P_{\rm F}$, and miss probability $P_{\rm M}$ are defined, but no detection probability of BKG class is defined because of its un-interestingness.

With the above interpretation, the probabilities of hypotheses can be considered as the significance of their classes. Specifically, $P_{\rm D}$ can be translated to the correct classification rate of a small but significant signal class as accuracy of signal class, $P_{\rm A}$, which can be particularly taken care of by $P_{\rm D}$. On the other hand, $P_{\rm F}$ can be translated to MC, $P_{\rm MC}$, to indicate BKG sources specified by H_0 that are falsely classified into the signal class specified by H_1. As a result, the commonly used evaluation performance tool for SD, ROC analysis can be readily applied to a classifier, where an ROC curve is plotted as a function of $P_{\rm A}$ versus false alarm probability, $P_{\rm MC}$.

Table I can be reinterpreted as SD, false signal (FS) detection, false BKG (FB) detection, and BKG detection (BD), respectively, which produces a signal-BKG classification confusion matrix in Table II. This is the Table that can be used to address the two issues, BKG and unbalanced class.

Since pdfs in (7) are generally not available in real applications, the confusion matrix in Table II must be obtained by a two-class confusion matrix shown in Fig. 4.

Using Fig. 4, all the necessary definitions can be defined as follows.

$n_{\rm s}$ = the number of signal samples;

$n_{\rm SD}$ = the number of detected signal samples;

$n_{\rm BKG}$ = the number of not signal samples corresponding to BKG samples;

$n_{\rm BD}$ = the number of BKG samples detected as BKG samples;

$n_{\rm FS}$ = the number of BKG samples detected as signal samples;

$n_{\rm FB}$ = the number of signal samples detected as BKG samples;

N = the total number of data samples $= n_{\rm s} + n_{\rm BKG}$

$$P_{\text{detection power/probability}} = P_{\rm D} = p(\text{SD}|\text{true signal present}) = \frac{n_{\rm SD}}{n_{\rm SD} + n_{\rm FB}} \tag{7}$$

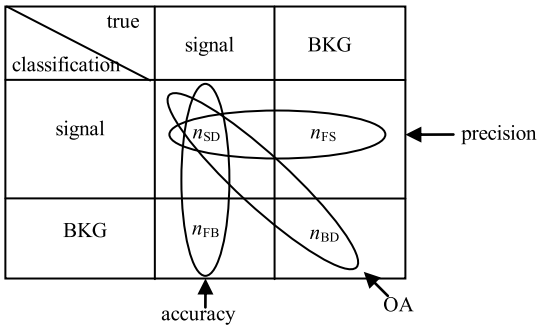


Fig. 4. Signal-BKG two class confusion matrix.

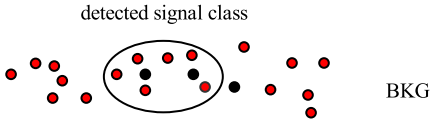


Fig. 5. Simple example of SD in BKG.

$$P_{\text{miss probability}} = P_{\rm M} = 1 - P_{\rm D} \tag{8}$$

$$P_{\text{false alarm probability}} = P_{\rm F} = \frac{n_{\rm FS}}{n_{\rm BKG}} = \frac{n_{\rm FS}}{n_{\rm FS} + n_{\rm BD}} \tag{9}$$

$$P_{\text{precision}} = P_{\rm PR} = p(\text{SD}|\text{detector}) = \frac{n_{\rm SD}}{n_{\rm SD} + n_{\rm FS}} \tag{10}$$

$$P_{\text{accuracy}} = P_{\rm A} = P_{\rm D} = p(\text{SD}|\text{data}) \tag{11}$$

$$P_{\text{overall accuracy}} = P_{\rm OA} = p(\text{SD}|\text{data}) = \frac{1}{N}(n_{\rm SD} + n_{\rm BKG}). \tag{12}$$

As a simple example to illustrate the imbalanced class and BKG issues, assume that a small-signal class contains only three data samples highlighted by the black circles and the BKG class contains 18 data samples highlighted by the red circles, as shown in Fig. 5.

Furthermore, assume that two signal samples and six BKG samples are detected as signals in the detected signal class and one signal sample detected as a BKG samples in the detected BKG class with 12 BKG sample correctly detected as BKG samples. In this case, a class is considered as a desired signal to be detected in BKG regardless of whether a class is small or large so that the imbalanced class and BKG issues can be addressed by OCD as SD in BKG. Table III tabulates the results in Fig. 5 as a confusion matrix with $P_{\rm PR} = 2/8$, $P_{\rm D} = P_{\rm A} = 2/3$, $P_{\rm F} = 6/8$, and $P_{\rm OA} = 14/21$.

It is worth noting that there is a significant difference between detection and binary classification. A detector performs SD by finding an appropriate threshold τ which corresponds to finding a boundary between two classes found by binary classification. By contrast, a binary classifier performs

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TABLE III
CONFUSION MATRIX FOR SIGNAL-BKG TWO-CLASS CLASSIFICATION FOR THE EXAMPLE IN FIG. 5

classification \ true	signal class	BKG class
detected signal class	2	6
detected BKG class	1	12

classification by assigning each data sample to one of the two classes. It does not find the boundary determined by the threshold τ as the way a detector does. In light of image processing, a detector finds edges in the data to segment foreground from the BKG, whereas a binary classifier segments the image data into foreground and BKG, each of which represents one single class. With this interpretation, the OCD discussed above can also be considered as an OCCB without modeling BKG as opposed to the OCC in [43]–[54]. In other words, when OCD is applied to classification problems, it is referred to as OCCB.

IV. MULTIHYPOTHESES TESTING

Theoretically speaking, a single target detector using binary hypothesis testing problem (1) can be simply extended to a multiple-target detector using multiple-hypotheses testing problem by expanding one single target to M targets.

A. Multitarget Detection

Assume that there are M signals and each signal is specified by one hypothesis as follows:

$$\begin{aligned} &H_0 : s_1 \approx p_0(y) \\ &\vdots \\ &H_M : s_M \approx p_M(y) \end{aligned} \tag{13}$$

where $p_j(y) = p(y|H_j)$ is the pdf to describe the jth hypothesis, H_j. Generally, there are two ways to perform an M-hypotheses testing problem [42].

1) One-Against-All Approach: One common approach uses the one-against-all strategy which finds the hypothesis with the largest *a posteriori* probability $p_{j^*}(y)$ satisfying

$$j^*(y) = \arg\left\{ \max_{1\le j\le M} p_j(y) \right\} \tag{14}$$

in which case y is assigned to the $j*$ signal. The detector resulting from (14) is called MAP detector and can be referred to as winner-take-all detector.

2) One-Against-One Approach: Another approach uses one-to-one strategy, which runs through all possible pairs of binary hypotheses out of M hypotheses. Then, the SD is performed by a majority vote. More specifically, let $J = \binom{M}{2} = (M(M-1))/2$ be the number of pairs of binary hypotheses derived from M hypotheses and $n_j(y)$ be the number of binary hypothesis tests that declare the y observation to be the jth signal among the J pairs of binary hypothesis testing problems. That is, the observation y will be declared to be the $j*$ signal if it satisfies

$$j^*(y) = \arg\left\{ \max_{1\le j\le J} n_j(y) \right\}. \tag{15}$$

B. Multiclass Classification in BKG

Following a similar approach to signal classification in BKG problems in (6), we can also formulate a multitarget detection problem by including an extra hypothesis H_{BKG} particularly correspond to BKG as follows:

$$\begin{aligned} &H_1 : \text{class } 1 \approx p_1(y) \\ &\vdots \\ &H_M : \text{class } M \approx p_M(y) \\ &H_{\text{BKG}} : \text{BKG class} \approx p_{\text{BKG}}(y) \end{aligned} \tag{16}$$

where $p_j(y) = p(y|H_j)$ is the pdf of the observation y under hypothesis H_j to determine y belonging to the jth class H_j. The resulting classifier can be considered as MCCB. The key difference between OCCB and MCCB is that OCCB classifies one class at a time when it is applied to classifying multiple classes and MCCB classifies multiple classes all together in a shot operation.

Since there is no availability of pdfs in (16), we can use Fig. 4 to convert an M-target detection in BKG to an M classes-plus-BKG confusion matrix constructed in Fig. 6, where each target corresponds to one particular class with all definitions summarized as follows.

M = the number of classes, each of which corresponds to one target;

j = index to indicate a data sample in the jth class according to ground truth;

i = index to indicate the ith classified sample;

C_j = the set of data samples in the jth class;

n_{ij} = the number of signal samples in the jth class C_j to be classified into the ith class C_i;

n_j = the number of data samples in the jth class C_j, i.e., $n_j = \sum_{i=1}^{M} n_{ij}$;

N = the total number of data samples, $N = \sum_{j=1}^{M} n_j$;

n_{jj} = the number of signal samples in the jth class correctly classified into the jth class;

$\hat{n}_{ij}$ = the number of data samples classified in the ith class, C_i, which are supposed to in the jth class, C_j. In this case, $\hat{n}_{ij} = n_{ij}$;

$\hat{C}_i$ = the set of data samples being classified in the ith class, C_i;

$\hat{n}_i$ = the number of data samples classified by $\delta(\mathbf{r})$ in the ith class, $\hat{C}_i$, i.e., $\hat{n}_i = \sum_{j=1}^{M} \hat{n}_{ij}$;

$\tilde{n}_j = \sum_{i=1, i\neq j}^{M} n_{ij}$ = the number of signal samples in the jth class misclassified into the ith class with $i \neq j$;

$\tilde{C}_i$ = the set of data samples in the ith class which are misclassified into the classes $\{C_j\}_{j=1, j\neq i}^{M}$

$$p(C_j) = \text{sample ratio} = \frac{n_j}{N} \tag{17}$$

$$\begin{aligned} P_{\text{A}}(C_j) &= \text{accuracy of the } j\text{th class} \\ &= p(C_{jj}|C_j) = \frac{n_{jj}}{\sum_{i=1}^{M} n_{ij}} \end{aligned} \tag{18}$$

$$\begin{aligned} P_{\text{OA}} &= p_{\text{overall-accuracy}}\left(\{C_j\}_{j=1}^{M} \right) \\ &= \sum_{j=1}^{M} p(C_j)\, p(C_{jj}|C_j) \end{aligned}$$

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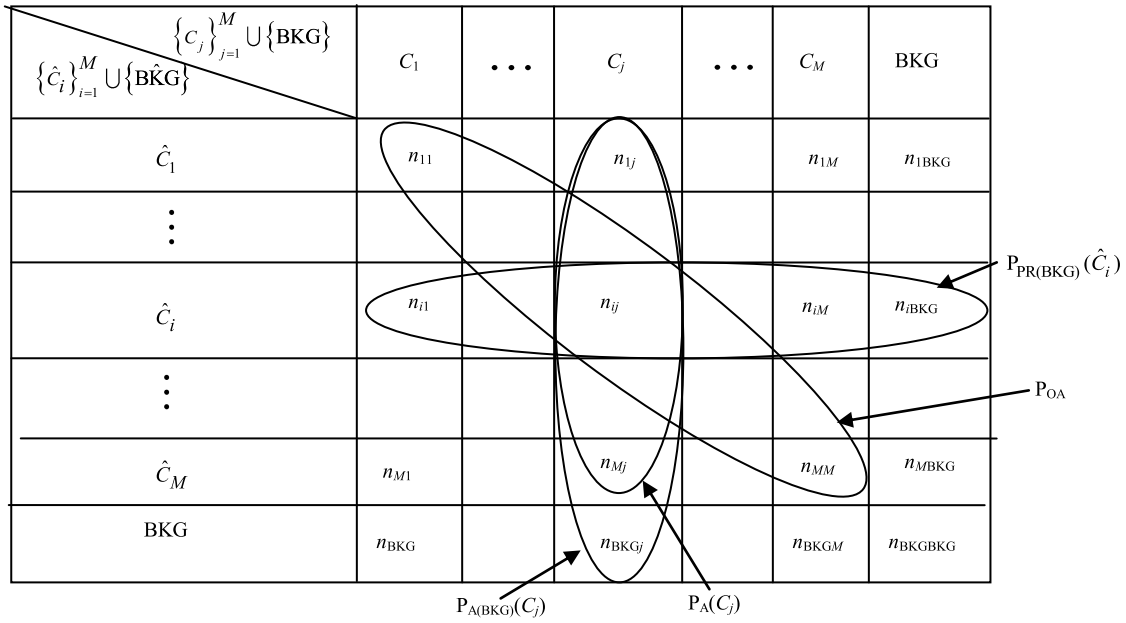


Fig. 6. P_A, $P_{A(BKG)}$, $P_{PR(BKG)}$, and P_{OA} calculated M-class confusion matrix along with BKG class to be considered as a single separate class.

$$= \frac{1}{N}\sum_{j=1}^{M} n_j \left(\frac{n_{jj}}{\sum_{i=1}^{M} n_{ij}}\right) = \frac{1}{N}\sum_{j=1}^{M} n_{jj} \quad (19)$$

$$P_{AA} = (1/M)\sum_{j=1}^{M} n_{jj}/n_j \quad (20)$$

$$p(\tilde{C}_j|C_j) = \text{misclassification rate of the } j \text{ th class}, C_j$$
$$= \sum_{i=1, i\neq j}^{M} p(C_{ij}|C_j) = 1 - p(C_{jj}|C_j)$$
$$= 1 - P_A(C_j) \quad (21)$$

$$P_{MC}(\tilde{C}) = \text{misclassification rate}$$
$$= \sum_{j=1}^{M} p(C_j)\, p(\tilde{C}_j|C_j) \quad (22)$$

$$P_{\text{precision(BKG)}}(\hat{C}_i) = P_{PR(BKG)}(\hat{C}_i)$$
$$= p(\hat{C}_i|\text{classification})$$
$$= \frac{\hat{n}_{ii}}{\sum_{j=1}^{M} \hat{n}_{ij} + \hat{n}_{\text{BKG}}} \quad (23)$$

$$P_{OPR} = \sum_{i=1}^{M} \frac{\hat{n}_i}{N} P_{PR(BKG)}(\hat{C}_i)$$
$$= \frac{1}{N}\sum_{i=1}^{M} \hat{n}_i P_{PR(BKG)}(\hat{C}_i) \quad (24)$$

$$P_{A(BKG)}(C_j) = n_{jj}/(n_j + n_{\text{BKG}}) \quad (25)$$

$$P_{OA(BKG)} = \sum_{j=1}^{M} p(C_j)\, P_{A(BKG)}(C_j). \quad (26)$$

Fig. 7 shows a two-class classification simple example to illustrate the FCMA issue where a class C_1 contains nine data samples highlighted by three open circles and six black solid circles and another class C_2 which has 11 data samples highlighted by three open circles and eight red solid circles with three data samples in common with C_1 highlighted by open circles. In this case, C_1 and C_2 share the same three data samples highlighted by three open circles. This scenario occurs when a soft-decision classifier performs FCMA as a detector similar to (3) by using a threshold τ. In addition, nine BKG pixels are highlighted by yellow solid circles. In Fig. 7, $\hat{C}_1$ contains 7 data samples in C_1, 2 data samples in C_2, and 2 BKG samples with a total of 11 data samples, i.e., $\hat{n}_1 = 9 + 2(\text{BKG})$, while $\hat{C}_2$ contains eight data samples in C_2, one data samples in C_1, and one BKG sample with a total of ten data samples, i.e., $\hat{n}_2 = 9 + 1(\text{BKG})$, where two BKG samples are misclassified into $\hat{C}_1$ and one BKG sample is misclassified into $\hat{C}_2$.

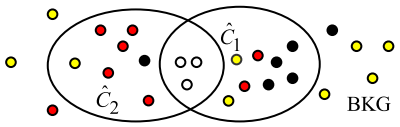


Fig. 7. Simple example of illustrating the FCMA issue.

TABLE IV
CONFUSION MATRIX FOR THE EXAMPLE IN FIG. 7

classification \ true	class C_1	class C_2	BKG
class $\hat{C}_1$	7	2	2
class $\hat{C}_2$	1	8	1
BKG	1	1	6

In this case, $n_1 = 9$, $n_2 = 11$, $n_{11} = 7$, $n_{22} = 8$, $n_{\text{BKGBKG}} = 6$, $n_{21} = 1$, $\hat{n}_{21} = 1$, $n_{12} = 2$, $\hat{n}_{12} = 2$, $n_{1\text{BKG}} = 2$, $n_{2\text{BKG}} = 1$, $\hat{n}_{\text{BKG1}} = 1$, and $\hat{n}_{\text{BKG2}} = 1$. Table IV constructs the confusion matrix of the example in Fig. 7 along with their corresponding accuracy and PR.

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V. Effects of BKG on Classification

In HSIC, the classes of interest are assumed to make up the entire data samples, where P_{OA} and P_{A} are only defined on classes of interest by assuming that the BKG class is of no interest and excluded from classification. However, due to significant high spectral resolution, many unknown data samples can be uncovered by a hyperspectral imaging sensor but cannot be labeled. In this case, when it comes to HSIC, these unlabeled data samples are generally grouped into a BKG class, which is considered as an uninteresting class. Consequently, such a BKG class is usually excluded from HSIC. Unfortunately, on many occasions, the BKG class plays an integral part of the data where many BKG data samples are very easily misclassified into various classes of interest and cause MC errors, which can be considered as false alarm probabilities. Discarding or ignoring the BKG class may result in misleading conclusions. To address this issue, the OCC developed in [43]–[54] attempted by grouping all uninteresting classes to form a BKG class [53]. However, including the real BKG class for classification turns out to be a very difficult task and seems not being investigated.

Most interestingly, it is the BKG class that attributes to MC. Such an issue has been overlooked in supervised HSIC. Compared to P_{OA} and P_{A}, P_{PR} in (23) has taken care of the effect of BKG class on other classes of interest since it is defined based on the classified data samples where the classified data samples may be very likely to be misclassified into the BKG class but not into other classes since BKG generally plays a buffer among different disjoint classes [such as class 17 in the Purdue data in Fig. 10(b)]. Fig. 7 clearly explains how to factor in BKG into classification where the hypothesis H_{BKG} in (16) plays a role similar to H_0 representing noise in (1). It is obvious that traditional supervised classifiers cannot deal with the BKG issue since the provided ground truth already excludes BKG from classification.

This section presents two classifiers, which can take care of the BKG issue. One is a single target detector, called CEM [53]–[56], which can be considered as an OCCB and the other is multitarget detector, called LCMV [57], which can be considered as an MCCB [29].

Interestingly, to deal with the BKG issue for classification, a recent work in [41] develops a one-class approach, OCC, which used cluster space representation to model BKG to achieve better classification. Such an OCC shares a similar idea with target detection where the class of interest to be classified is considered as one hypothesis corresponding to the desired signal to be detected versus BKG as an uninteresting signal corresponding to another hypothesis to represent noise. Compared to OCC in [41], CEM and LCMV took a rather different approach by inverting the sample correlation matrix $\mathbf{R}$ to suppress BKG rather than modeling BKG, which requires tuning several parameters that have significant impact on classification. As a result, CEM and LCMV are much easily implemented than OCC since they only need the knowledge of the targets of interest to be detected and nothing else.

A. OCC in BKG: CEM

Suppose that a hyperspectral image is represented by a collection of image pixel vectors, denoted by $\{\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N\}$ where $\mathbf{r}_i = (r_{i1}, r_{i2}, \ldots, r_{iL})^{\mathrm{T}}$ for $1 \leq i \leq N$ is an L-dimensional pixel vector, N is the total number of pixels in the image, and L is the total number of spectral channels. Furthermore, assume that $\mathbf{d} = (d_1, d_2, \ldots, d_L)^{\mathrm{T}}$ is specified by a desired signature of interest to be used for target detection. The CEM developed in [53] is particularly designed to detect target signal sources specified by $\mathbf{d}$ while minimizing the energy caused by the unknown signal sources via a finite impulse response (FIR) linear filter specified by L filter coefficients, $\{w_1, w_2, \ldots, w_L\}$, denoted by an L-dimensional vector $\mathbf{w} = (w_1, w_2, \ldots, w_L)^{\mathrm{T}}$ subject to the constraint $\mathbf{d}^{\mathrm{T}}\mathbf{w} = \mathbf{w}^{\mathrm{T}}\mathbf{d} = 1$. More specifically, let y_i denote the output of the designed FIR filter resulting from the input $\mathbf{r}_i$. Then, y_i can be expressed by

$$y_i = \sum_{l=1}^{L} w_l r_{il} = (\mathbf{w})^{\mathrm{T}}\mathbf{r}_i = \mathbf{r}_i^{\mathrm{T}}\mathbf{w} \tag{27}$$

and the average energy of the filter output is given by

$$\begin{aligned}(1/N)\sum_{i=1}^{N} y_i^2 &= (1/N)\sum_{i=1}^{N}\left(\mathbf{r}_i^{\mathrm{T}}\mathbf{w}\right)^2 \\ &= \mathbf{w}^{\mathrm{T}}\left[(1/N)\sum_{i=1}^{N}\mathbf{r}_i\mathbf{r}_i^{\mathrm{T}}\right]\mathbf{w} = \mathbf{w}^{\mathrm{T}}\mathbf{R}_{L\times L}\mathbf{w}\end{aligned} \tag{28}$$

where $\mathbf{R} = (1/N)\sum_{i=1}^{N}\mathbf{r}_i\mathbf{r}_i^{\mathrm{T}}$ is the sample autocorrelation matrix of the image. The goal is to solve the following linearly constrained optimization problem:

$$\begin{aligned}&\min_{\mathbf{w}}\left\{\mathbf{w}^{\mathrm{T}}\mathbf{R}\mathbf{w}\right\} \\ &\text{s.t. } \mathbf{d}^{\mathrm{T}}\mathbf{w} = \mathbf{w}^{\mathrm{T}}\mathbf{d} = 1\end{aligned} \tag{29}$$

where $\mathbf{w}^T\mathbf{R}\mathbf{w}$ can be considered as the variance resulting from signals not passing through the filter. The optimal solution to (29) is shown in [53] to be

$$\mathbf{w}^{\mathrm{CEM}} = \left(\mathbf{d}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{d}\right)^{-1}\mathbf{R}^{-1}\mathbf{d} \tag{30}$$

and a CEM detector, $\delta^{\mathrm{CEM}}(\mathbf{r})$, is specified by

$$\delta^{\mathrm{CEM}}(\mathbf{r}) = \left(\mathbf{w}^{\mathrm{CEM}}\right)^{\mathrm{T}}\mathbf{r} = \left(\mathbf{d}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{d}\right)^{-1}\left(\mathbf{R}^{-1}\mathbf{d}\right)^{\mathrm{T}}\mathbf{r}. \tag{31}$$

If the BKG is made up of all unknown signal sources, then the NPD $\delta^{\mathrm{NP}}(\mathbf{r})$ in (3) is replaced by $\delta^{\mathrm{CEM}}(\mathbf{r})$ in (31) with

$$\begin{aligned}\Lambda(\mathbf{r}) &= \left(\mathbf{d}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{d}\right)^{-1}\left(\mathbf{R}^{-1}\mathbf{d}\right)^{\mathrm{T}}\mathbf{r} \\ &= \left(\mathbf{d}^{T}\mathbf{R}^{-1}\mathbf{d}\right)^{-1}\mathbf{d}^{\mathrm{T}}\left(\mathbf{R}^{-1}\mathbf{r}\right)\end{aligned} \tag{32}$$

for the binary hypothesis testing problem exactly specified by (6), where $\mathbf{R}^{-1}\,\mathbf{r}$ in (32) suppresses the BKG while the signal being detected by the matching signal $\mathbf{d}$ via a match filter $\mathbf{d}^T\tilde{\mathbf{r}}$ where $\tilde{\mathbf{r}} = \mathbf{R}^{-1}\mathbf{r}$ is the BKG -suppressed $\mathbf{r}$. Now, we can interpret the signal of interest, $\mathbf{d}$, as class mean, μ_i, of a single class of interest, C_i, and then, the $\delta^{\mathrm{CEM}}(\mathbf{r})$ in (31) can be used as OCCB to classify a specific class, C_i in BKG.

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B. Multiclass Classification in BKG: LCMV

LCMV extends CEM as an OCCB to an MCCB. Suppose that $\mathbf{d}_1, \mathbf{d}_2 \cdots \mathbf{d}_M$ are M specific class signatures of interest, which can be either provided by *a priori* knowledge or obtained by training samples.

Now, we interpret each of $\mathbf{d}_1, \mathbf{d}_2 \cdots \mathbf{d}_M$ as a desired signal arrival direction in adaptive beamforming in [57] and let $\mathbf{D}^M = [\mathbf{d}_1 \mathbf{d}_2 \cdots \mathbf{d}_M]$ be an $L \times M$ class signature matrix, where L is the total number of band. Assume that an FIR linear filter is the same one implemented in CEM. Then, the LCMV problem considered in [57] can be reformulated as the following M-class constrained optimization problem:

$$\begin{aligned} &\min_{\mathbf{w}} \left\{ \mathbf{w}^T \mathbf{R} \mathbf{w} \right\} \\ &\text{s.t. } \left(\mathbf{D}^M \right)^T \mathbf{w} = \mathbf{1}^M \end{aligned} \tag{33}$$

where $\mathbf{1}^M = (1_1, 1_2, \ldots, 1_M)^T$ is an M-dimensional constraint unity vector and each of M constraints, $1_1, 1_2, \ldots, 1_M$ is used imposed on a particular class, for example, the jth constraint, 1_j is imposed on the jth class. The optimal solution to (33) can be solved by

$$\mathbf{w}^{\mathrm{LCMV}} = \mathbf{R}^{-1} \mathbf{D}^M \left(\left(\mathbf{D}^M \right)^T \mathbf{R}^{-1} \mathbf{D}^M \right)^{-1} \mathbf{1}^M. \tag{34}$$

where $(\mathbf{D}^M)^T \mathbf{R}^{-1} \mathbf{D}^M$ is an $M \times M$ matrix and $\mathbf{R} = (1/N) \sum_{i=1}^{N} \mathbf{r}_i \mathbf{r}_i^T$ is the sample correlation matrix of size $L \times L$. An approach using (33) to derive the optimal solution (34) is called LCMV.

By virtue of (34), we can define a linear classifier by

$$\delta^{\mathrm{LCMV}}(\mathbf{r}) = \left(\mathbf{w}^{\mathrm{LCMV}} \right)^T \mathbf{r} \tag{35}$$

which results in an LCMV classifier with M-class signature matrix $\mathbf{D}^M$ constrained by $\mathbf{1}^M$ and BKG-suppressed by $\mathbf{R}^{-1}$.

In analogy with CEM, if the BKG is made up of all unknown signal sources, then the NPD $\delta^{\mathrm{NP}}(\mathbf{r})$ in (3) is replaced by $\delta^{\mathrm{LCMV}}(\mathbf{r})$ in (35) with

$$\Lambda(\mathbf{r}) = \left(\mathbf{R}^{-1} \mathbf{D}^M \left(\left(\mathbf{D}^M \right)^T \mathbf{R}^{-1} \mathbf{D}^M \right)^{-1} \mathbf{1}^M \right)^T \mathbf{r} \tag{36}$$

for the multiple hypothesis testing problem exactly specified by (16). In this particular case, $\mathbf{R}^{-1}$ and the signal vector $\mathbf{D}^M$ are combined to form a $((\mathbf{D}^M)^T \mathbf{R}^{-1} \mathbf{D}^M)^{-1}$-suppressed and signal matched operator, $\mathbf{R}^{-1} \mathbf{D}^M ((\mathbf{D}^M)^T \mathbf{R}^{-1} \mathbf{D}^M)^{-1} \mathbf{1}^M$, to perform a matched filter in a similar manner to CEM.

Now, combining (6) and (16) yields a multiclass classification problem by the following binary hypothesis testing problem:

$$\begin{aligned} &H_0 : \text{BKG class} \\ &\text{versus} \\ &H_1 : \text{multiple classes } \{C_1, C_2, \ldots, C_M\}. \end{aligned} \tag{37}$$

In this case, $\mathbf{D}^M$ in (33) is specified by M class means, $\{\boldsymbol{\mu}_j\}_{j=1}^M$, to produce $[\boldsymbol{\mu}_1 \boldsymbol{\mu}_2 \cdots \boldsymbol{\mu}_M]$ as class mean matrix for LCMV to be used $\mathbf{D}^M$ in (33) to perform MCCB by simultaneously classifying M classes in BKG.

VI. 3-D ROC Analysis for Classification

As noted in Section IV, both CEM and LCMV use binary and multihypothesis testing problems (6) and (16) to perform like OCCB and MCCB. Therefore, the classification of CEM and LCMV can be evaluated by ROC analysis. According to NPD, its performance is determined by the significance of level, β, which is specified by P_{F}, and then, β in turn determines P_{D}. However, in fact, β is actually determined by the threshold τ, which corresponds to the unknown cost incurred by a detector. As a consequence, both P_{D} and P_{F} are indeed determined by the unknown threshold τ, more specifically, functions of τ. A 3-D ROC curve simply includes τ as an independent variables and P_{D}, P_{F} as dependent variables through τ in a 3-D space of $(P_{\mathrm{D}}, P_{\mathrm{F}}, \tau)$. Using 3-D ROC analysis, a one-class classifier can be basically considered as a detector specified by a particular τ, which determines the hyperplane to produce a single point of $(P_{\mathrm{D}}, P_{\mathrm{F}})$.

In traditional supervised HSIC, such an ROC analysis is not applicable because each classifier is a hard-decision maker and can only generate a single pair of P_{D} and P_{F}, $(P_{\mathrm{D}}, P_{\mathrm{F}})$ as one point on an ROC curve. Hence, a common measure to evaluate classification performance is OA. This section presents an alternative ROC-based evaluation tool to measure the class accuracy, BKG suppression capability, and effectiveness of a classifier.

From the hypothesis testing problem specified by (1), a detector makes a binary hard decision by thresholding a real-valued LRT, $\Lambda(\mathbf{r})$ via a threshold τ in (3). Accordingly, the detector performance is actually determined by two parameters, the $\Lambda(\mathbf{r})$ and the τ, both of which are real values. As a result, the detection rate, P_{D}, in (4) and the false alarm probability/rate, P_{F}, in (5) are functions of the $\Lambda(\mathbf{r})$ and the threshold τ.

By virtue of (4) and (5), we consider the threshold parameter τ as an independent variable, and P_{D} and P_{F} as dependent variables through (4) and (5). The 3-D ROC analysis extends the traditional 2-D ROC analysis by including the threshold τ as a third dimension in addition to the 2-D specified by the parameters P_{D} and P_{F} used by the 2-D ROC analysis. In other words, the proposed 3-D ROC analysis produces a 3-D ROC curve plotted based on three parameters, P_{D}, P_{F}, and τ, referred to as 3-D ROC curve of $(P_{\mathrm{D}}, P_{\mathrm{F}}, \tau)$ from which three new 2-D ROC curves can also be derived, referred to as 2-D ROC curve of $(P_{\mathrm{D}}, P_{\mathrm{F}})$, 2-D ROC curve of (P_{D}, τ), and 2-D ROC curve of (P_{F}, τ), where the 2-D ROC curve of $(P_{\mathrm{D}}, P_{\mathrm{F}})$ turns out to be the traditional ROC curve.

Procedure for 3D ROC Curve Generation

1. Let $\mathbf{r}$ and $\delta(\mathbf{r})$ be a data sample vector and a soft decision (i.e., real-valued) classifier operating on $\mathbf{r}$ respectively. It should be noted that the $\delta(\mathbf{r})$ can be also interpreted as FCMA classifier discussed in Section VI such as edge preserving filtering (EPF)-based methods in [23].
2. Derive a normalized classifier $\delta_{\mathrm{normalized}}(\mathbf{r})$ from $\delta(\mathbf{r})$ as

$$\delta_{\mathrm{normalized}}(\mathbf{r}) = \frac{\delta(\mathbf{r}) - \min_{\mathbf{r}} \delta(\mathbf{r})}{\max_{\mathbf{r}} \delta(\mathbf{r}) - \min_{\mathbf{r}} \delta(\mathbf{r})}. \tag{38}$$

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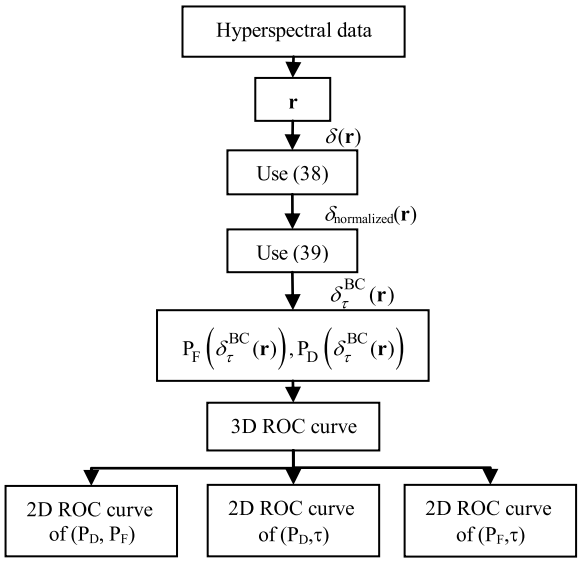


Fig. 8. Diagram of generating a 3-D ROC curve.

3. Let τ be the threshold which varies between [0,1] and introduce a binary-valued classifier (BC), $\delta_\tau^{\text{BC}}(\mathbf{r})$, resulting from $\delta_{\text{normalized}}(\mathbf{r})$ in (38) by

$$\delta_\tau^{\text{BC}}(\mathbf{r}) = \begin{cases} 1, & \text{if } \delta_{\text{normalized}}(\mathbf{r}) \geq \tau \\ 0, & \text{if } \delta_{\text{normalized}}(\mathbf{r}) < \tau \end{cases} \tag{39}$$

4. Calculate $P_\text{D}(\delta_\tau^{\text{BC}}(\mathbf{r}))$ and $P_\text{F}(\delta_\tau^{\text{BC}}(\mathbf{r}))$.
5. Plot a 3D ROC curve of (P_D,P_F, τ). (It should be noted that each threshold τ produces a point on the 3D ROC curve.

Fig. 8 depicts a flowchart of how to generate a 3-D ROC curve.

To evaluate the effectiveness of a detector quantitatively, AUC is generally used for this purpose. As a result, the higher the AUC of a 2-D ROC curve of (P_D, P_F) is, the better the detector is. Similarly, the AUC of a 2-D ROC curve of (P_D, τ) can be used to measure the class accuracy, that is, the higher the AUC of an 2-D ROC curve of (P_D, τ) is, the better the detector power is. On the other hand, if we consider all signal sources other than the target signal of interest as BKG, then the AUC of a 2-D ROC curve of (P_F, τ) can be further used to assess the BKG suppression impact on classification, viz., the smaller the AUC of an 2-D ROC curve of (P_F, τ) is, the better BKG suppression the detector has.

In addition to the advantages of using the 3-D ROC analysis described above, one particular goal of developing 3-D ROC analysis is to deal with classifiers which make soft decisions. In HSIC, a classifier usually includes a spatial filter to capture spatial information to improve classification. As a result, a spatial-filtered classifier is generally real-valued to make soft decisions, which indicate how much a probability or a likelihood a data sample is assigned to a class such as probability maps of classes in [23] or abundance fractions of a data sample present in a class such as CEM in (31) and LCMV in (35). In order to make hard decisions for class membership assignment, a thresholding technique is required, for example, MAP used in [30] and Otsu's method [58] used in ICEM [28] and ILCMV [29]. Two scenarios are of particular interest according to what type of a spectral classifier is used in HSIC.

1) *Pure Pixel-Based Spectral Classifiers:* One of the most widely used spectral classifiers in HSIC is SVM, which performs pure pixel spectral classification. Since SVM does not take into account spatial information, it is generally followed by a spatial filter to capture spatial correlation surrounding SVM-classified data samples. Such an SVM classifier coupled with a spatial filter produces real-valued classification maps for each of classes, which are likelihood maps, also referred to as soft-decision classification maps, of each data sample to be assigned to different classes. In this case, an SVM-spatial filter classifier can generate 3-D ROC curves by thresholding soft-decision classification maps via a threshold τ into hard-decision classification maps as will be demonstrated by experiments in Section VII. To make hard decisions, these soft-decision maps must be converted to binary maps. There are two ways to do so. One is MAP that assigns a data sample to a class with its largest probability or maximum likelihood. The other is to use a thresholding technique, such as Otsu's method. The key difference between these two is that MAP always assigns each data sample to a unique class, while a thresholding technique may result in multiple-class or no-class membership assignment. As an illustrative example, the EPF probability maps developed in [23] are indeed soft-decision maps, and MAP is a soft-to-hard decision converter. In this case, the binarization is performed by MAP. As an alternative, we can also use Otsu's method to threshold the EPF soft-decision maps by the threshold τ specified by $\tau_{\text{Otsu}}^{\text{EPF}}$ to produce binary maps for classification.
2) *Mixed Pixel-Based Spectral Classifiers:* In contrast to SVM, we can also use CEM in (31) and LCMV in (35) to perform mixed pixel spectral classification, where the $\Lambda(\mathbf{r})$ in (3) is interpreted as a detector specified by either $\delta^{\text{CEM}}(\mathbf{r})$ or $\delta^{\text{LCMV}}(\mathbf{r})$ with the τ in (3) being used a parameter to threshold CEM/LCMV abundance fractional maps by varying its values to produce 3-D ROC curves. By virtue of 3-D ROC analysis, classification performance of CEM and LCMV can be measured via $\Lambda(\mathbf{r})$ specified by either $\delta^{\text{CEM}}(\mathbf{r})$ or $\delta^{\text{LCMV}}(\mathbf{r})$ by finding appropriate threshold value τ. The value produced by Otsu's method is only one of the values of τ that can be produced without specifying P_F.

VII. FCMA FOR HSIC

SS methods have been widely used in HSIC [6]–[15], [19]–[23]. When it comes to evaluation of classification performance, the confusion matrix in Fig. 4 has been used to calculate AA and OA, and unfortunately, it cannot be directly applicable to the FCMA

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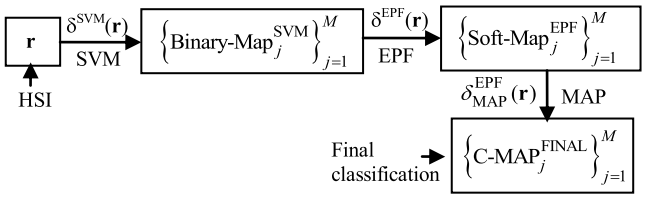


Fig. 9. Diagram of implementing an EPF-based method.

issue. This section addresses this issue to make ROC analysis also available to SS methods. In general, an SS method implements a spectral classifier for pure pixel classification followed by a spatial filter to capture spatial information from spectral-classification maps. By doing so, a recently developed approach, called EPF-based methods in [23], is used as an example to illustrate the proposed idea. It has been shown in [23] through a comprehensive comparative analysis that EPF-based methods performed better than most existing SS methods. There are four EPF-based techniques, EPF-B-c, EPF-G-c, EPF-B-g, and EPF-G-g, where "B" and "G" are used to specify a bilateral filter and a guided filter, respectively, and "g" and "c" indicate that the first principal component and the color composite of the three principal components are used as reference images [23].

The main idea of EPF-based methods depicted in Fig. 9 is to first use an SVM to perform spectral classification to produce binary maps for each of classes, $\{\text{Binary-MAP}_j^{\text{SVM}}\}_{j=1}^{M}$, and then apply an EPF such as guided or bilateral filter to capture spatial information from each of SVM-classified binary maps to produce final soft probability maps, $\{\text{Soft-MAP}_j^{\text{EPF}}\}_{j=1}^{M}$. Finally, the MAP is implemented to make hard decisions for final classification, $\{\text{C-MAP}_j^{\text{FINAL}}\}_{j=1}^{M}$.

Interestingly, it is the final soft probability maps, $\{\text{Soft-MAP}_j^{\text{EPF}}\}_{j=1}^{M}$, produced by an EPF that is actually an FCMA classifier which can make 3-D ROC analysis work. More specifically, we can define a probability vector for each data sample vector

$$\delta^{\text{EPF}}(\mathbf{r}) = \left(\delta_1^{\text{EPF}}(\mathbf{r}), \ldots, \delta_j^{\text{EPF}}(\mathbf{r}), \ldots, \delta_M^{\text{EPF}}(\mathbf{r})\right)^T \tag{40}$$

where

$$\delta_j^{\text{EPF}}(\mathbf{r}) = \text{Soft-Map}_j^{\text{EPF}}(\mathbf{r}). \tag{41}$$

The $\delta^{\text{EPF}}(\mathbf{r})$ in (40) plays the exactly the same role of $\delta^{\text{CEM}}(\mathbf{r})$ in (31) and $\delta^{\text{LCMV}}(\mathbf{r})$ in (35) with the only key difference in that $\delta^{\text{EPF}}(\mathbf{r})$ represents a probability vector as opposed to $\delta^{\text{CEM}}(\mathbf{r})$ and $\delta^{\text{CEM}}(\mathbf{r})$ which are abundance vectors. A 3-D ROC curve can then be generated by using varying threshold values of τ to convert the FCMA classifier $\delta^{\text{EPF}}(\mathbf{r})$ in (40) to a hard-decision maker $\delta_\tau^{\text{EPF}}(\mathbf{r})$.

It should be noted that once the threshold τ is specifically determined, the soft probability maps, $\{\text{Soft-MAP}_j^{\text{EPF}}\}_{j=1}^{M}$, produced by $\delta^{\text{EPF}}(\mathbf{r})$ in (40) become binary maps, $\{\text{Binary-MAP}_{\tau,j}^{\text{EPF}}\}_{j=1}^{M}$, for final classification. This gives rise to a challenging issue in how to find an optimal binarization. In [23], the binarization is performed by MAP. In this case, $\delta^{\text{EPF}}(\mathbf{r})$ in (40) is reduced to $\delta_{\text{MAP}}^{\text{EPF}}(\mathbf{r})$ and the resulting classification maps become the final classification maps $\{\text{Binary-MAP}_{\text{MAP},j}^{\text{EPF}}\}_{j=1}^{M}$. Of course, the MAP-based binarization may not be optimal. An alternative approach is to use thresholding technique to find an appropriate threshold τ. One widely used thresholding method was developed by Otsu [58]. As a result, $\delta^{\text{EPF}}(\mathbf{r})$ in (40) becomes $\delta_{\text{Otsu}}^{\text{EPF}}(\mathbf{r})$, which uses Otsu-determined $\tau_{\text{Otsu}}^{\text{EPF}}(\mathbf{r})$ to make hard decisions for final classification, $\{\text{Binary-MAP}_{\text{Otsu},j}^{\text{EPF}}\}_{j=1}^{M}$. However, there is a crucial difference between using MAP and Otsu's method to make hard decisions. When MAP is used, each data sample will be assigned to a unique class, while Otsu's method may assign some data samples to multiple classes or no classes.

In summary

$$\begin{array}{ccccc} \delta^{\text{EPF}}(\mathbf{r}) & \rightarrow & \delta_\tau^{\text{EPF}}(\mathbf{r}) & \rightarrow & \delta_{\text{MAP}}^{\text{EPF}}(\mathbf{r}) \\ & & \downarrow & & \\ & & \delta_{\text{Otsu}}^{\text{EPF}}(\mathbf{r}) & & \end{array} \tag{42}$$

and

$$\delta^{\text{CEM/LCMV}}(\mathbf{r}) \rightarrow \delta_\tau^{\text{CEM/LCMV}}(\mathbf{r}) \rightarrow \delta_{\text{Otsu}}^{\text{CEM/LCMV}}(\mathbf{r}) \tag{43}$$

where $\delta_\tau^{\text{CEM/LCMV}}(\mathbf{r})$ is $\delta^{\text{CEM}}(\mathbf{r})$ in (31) and $\delta^{\text{LCMV}}(\mathbf{r})$ in (35) using a threshold τ to make hard decisions.

Interestingly, a recent work, which combines EPF-based methods with the iterative process used in ICEM/ILCMV to derive iterative versions of EPF-based methods, (IEPF)-based methods, can also be extended to $\delta_\tau^{\text{IEPF}}(\mathbf{r})$ as follows in the same way that δ_τ^{EPF} is derived from (40)–(42):

$$\begin{array}{ccccc} \delta^{\text{IEPF}}(\mathbf{r}) & \rightarrow & \delta_\tau^{\text{IEPF}}(\mathbf{r}) & \rightarrow & \delta_{\text{MAP}}^{\text{IEPF}}(\mathbf{r}). \\ & & \downarrow & & \\ & & \delta_{\text{Otsu}}^{\text{IEPF}}(\mathbf{r}) & & \end{array} \tag{44}$$

There are four versions of EPF, such as EPF-B-c, EPF-G-c, EPF-B-g, and EPF-G-g, developed in [23], among which EPF-G-g has shown to perform at least comparably or better than other three versions. In this case, EPF-G-g was selected as an EPF representative and its iterative counterpart, IEPF-G-g in [49] for the following experiments.

VIII. Real Image Experiments

Three real hyperspectral images available at website http://www.ehu.eus/ccwintco/index.php?title = Hyperspectral_Remote_Sensing_Scenes were used for experiments, such as Purdue University's Indiana Indian Pines, Salinas, and University of Pavia, Italy, each of which has its own unique feature characteristics worth being explored. In order to demonstrate the utility of 3-D ROC analysis in HSIC, ICEM, ILCMV, EPF-G-g, and IEPF-G-g are implemented for performance comparison. Specifically, EPF-G-g and IEPF-G-g were implemented as classifiers according to (40), (42), and (44) to produce 3-D ROC curves in comparison with ICEM and ILCMV.

A. Purdue Indiana Indian Pines

The first real hyperspectral image to be used for experiments is a well-known AVIRIS image scene, Purdue Indiana Indian Pine test site (Purdue's data) shown in Fig. 10(a) and its ground truth in Fig. 10(b) along with its 17 class maps in Fig. 10(c). Table V also tabulates all the specific types of 16 classes. It has a size of 145 × 145 pixel vectors taken from an area of mixed agriculture and forestry

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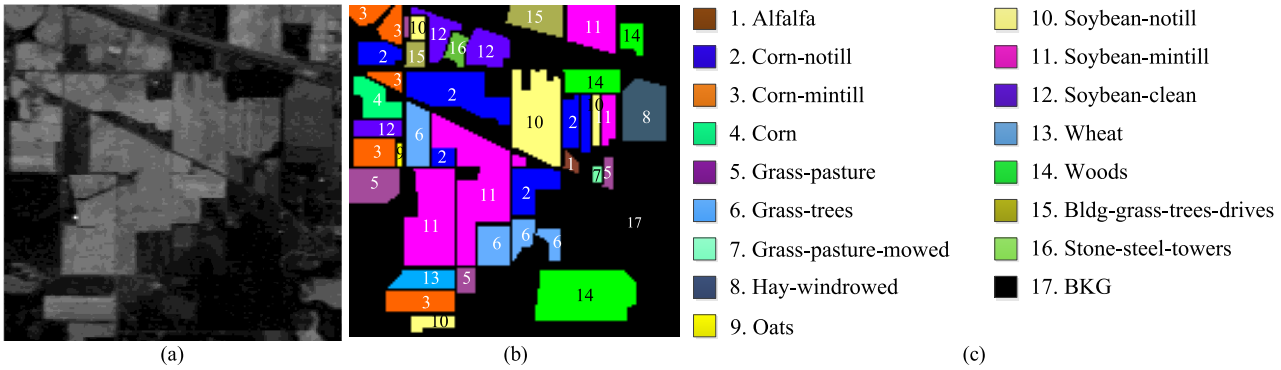


Fig. 10. Purdue's data and its ground truth with 16 classes. (a) Band 186 (2162.56 nm). (b) Ground truth map. (c) Classes by colors.

TABLE V
CLASS LABELS OF PURDUE'S DATA

class 1 (46)	Alfalfa	class 7 (28)	grass/pasture-mowed	class 13 (205)	wheat
class 2 (1428)	corn-notill	class 8 (478)	hay-windrowed	class 14 (1265)	woods
class 3 (830)	corn-min	class 9 (20)	oats	class 15 (386)	Bldg-grass green-drives
class 4 (237)	corn	class 10 (972)	soybeans-notill	class 16 (93)	stone-steel towers
class 5 (483)	grass/pasture	class 11 (2455)	soybeans-min	class 17 (10249)	BKG
class 6 (730)	grass/trees	class 12 (593)	soybeans-clean		

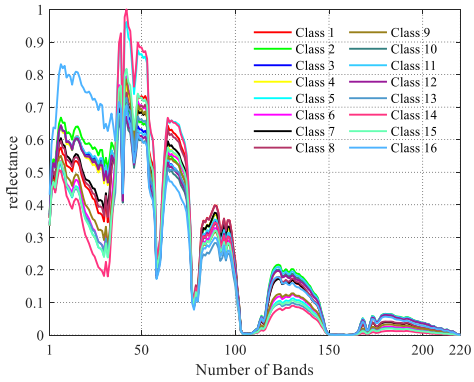


Fig. 11. Spectral profiles of 16-class sample means in Purdue's data.

in Northwestern Indiana, USA, with details of band and wavelength is given in caption. The data set is available at the website http://cobweb.ecn.purdue.edu/~biehl/MultiSpec/documentation.html. It was recorded in June 1992 with 220 bands with water absorption bands (bands 104–108 and 150–163, 220) removed.

Fig. 11 shows the spectral signatures of the 16 class means in Fig. 10(c). Fig. 12 shows 3-D ROC curves from the abundance fractional maps obtained by $\delta_\tau^{\text{EPF-G-g}}(\mathbf{r})$ in (42) along with their iterative counterparts $\delta_\tau^{\text{IEPF-G-g}}(\mathbf{r})$ in (44). Also, in Fig. 12, the 3-D ROC results produced by ICEM and ILCMV at the last iteration before their iterative processes were terminated. Specifically, three 2-D ROC curves of (P_D, P_F), (P_D, τ), and (P_F, τ) corresponding to their 3-D ROC curves are also plotted for 16 classes. The AUC values from three 2-D ROC curves in Fig. 12 were also calculated and shown as legends in Fig. 12. It is interesting to note that when 2-D ROC curves of (P_D, P_F) were plotted for EPF-G-g, $\delta_\tau^{\text{EPF-G-g}}(\mathbf{r})$ in (42) produces $P_F = 0$ when the threshold τ greater than a certain value $\beta < 1$ as shown by dashed vertical lines in Fig. 12. In this case, the AUC of 2-D ROC curves of (P_D, P_F) must be calculated by the area under the curve with P_F from 0 up to β and normalized by β. As we can see from Fig. 12, both of ICEM and ILCMV performed significantly better than EPF-based methods in terms of AUC values of all three types of 2-D ROC curves. In addition, comparing ILCMV to ICEM, ILCMV performed better than ICEM for classification of corn classes and soybean classes in the sense of two 2-D ROC curves of (P_D, τ) and (P_F, τ) but nearly the same for 2-D ROC curves of (P_D, P_F) which were all close to one. The results indicated that only examining traditional 2-D ROC curves of (P_D, P_F) was not sufficient to evaluate classification performance. The 3-D ROC analysis provides valuable information which allows users to evaluate classification performance based on P_D and P_F separately and individually via the AUC values calculated from 2-D ROC curves of (P_D, τ) and (P_F, τ).

In order to compare ROC results in Fig. 12, Table VI tabulates P_A, P_{MC}, and $P_{PR(BKG)}$ produced by EPF-G-g in [23] and IEPF-G-g developed in [49], where the results were calculated by ten runs using randomly selected training samples in the same way done in [23] with $\pm$ standard deviations. It should be noted that the standard deviation 0 in Table VI indicates that the values are smaller than 1%. In addition, Table VI also includes the results of P_A, P_{MC}, P_{AA}, P_{OA}, P_{OPR}, and $P_{PR(BKG)}$ produced by ICEM and ILCMV for comparison, where no $\pm$ standard deviations were calculated because ICEM and ILCMV used class means not randomly selected training samples as EPF-G-g and IEPF-G-g did. Since both ICEM and ILCMV are mixed pixel classifiers, they used Otsu's method to find appropriate threshold values $\tau_{\text{Otsu}}^{\text{CEM/LCMV}}(\mathbf{r})$ in (43) for classification. According to Table VI, we could conclude that IEPF-G-g was the best classifier followed

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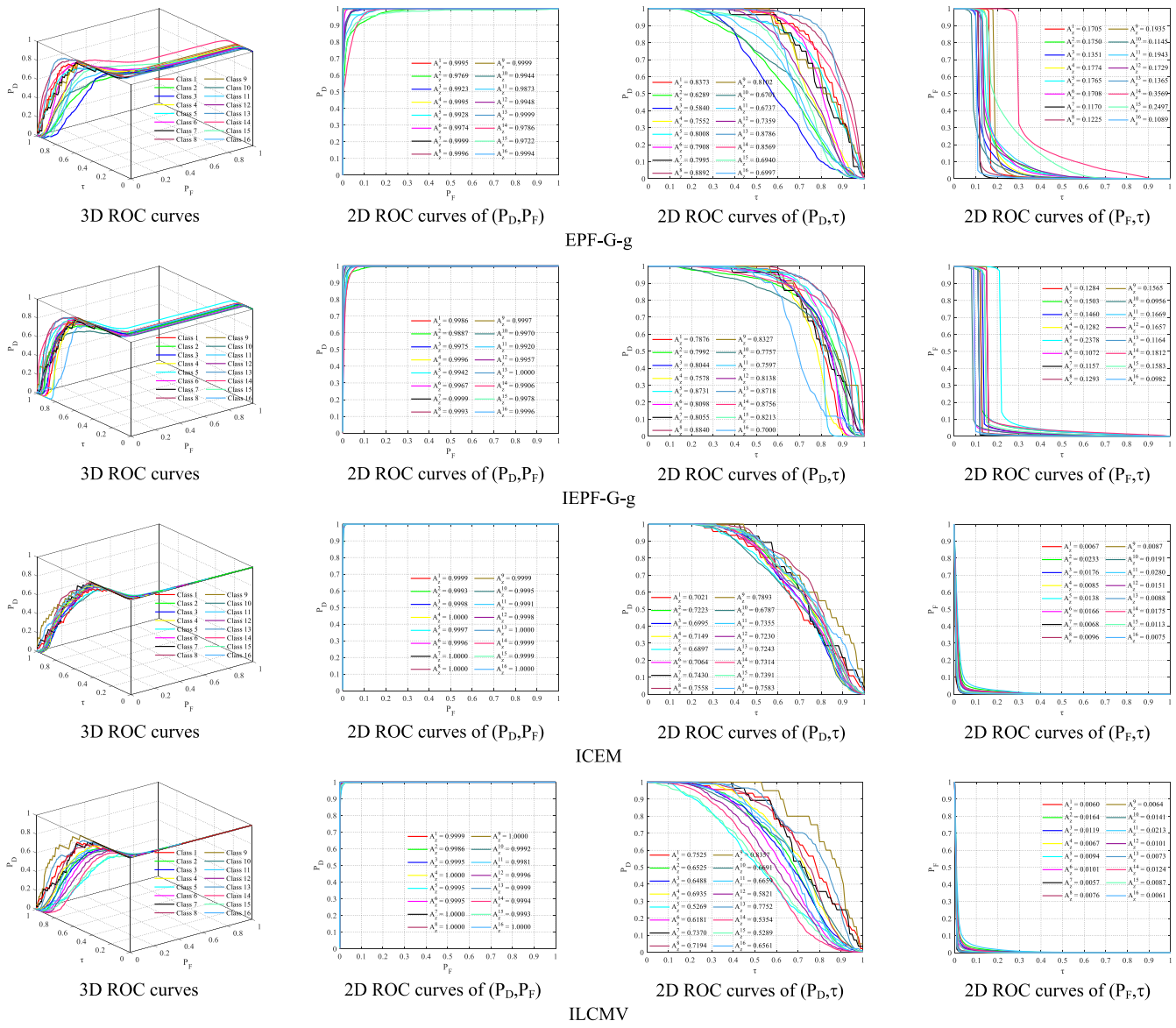


Fig. 12. 3-D ROC analysis produced by EPF-G-g, IEPF-G-g, ICEM, and ILCMV using Otsu's method for Purdue's data.

by ICEM and ILCMV with the worst results produced by EPF-G-g.

Comparing the results in Table VI with the results in Fig. 12, an immediate finding is that the AUC values of ROC curves of (P_D, τ) were not as good as P_A in Table VI. This is because the AUC values of (P_D, τ) were calculated as A_Z under an ROC curve of (P_D, τ), which is the integral of detection power/probability in BKG across the threshold τ from 0 to 1, whereas the P_A as class accuracy in Table VI was calculated without BKG. However, if we compare the AUC values of ROC curves of (P_D, τ) to $P_{PR(BKG)}$ in Table VI, their values are close since $P_{PR(BKG)}$ is calculated according to Fig. 4 which includes BKG data samples that were misclassified into the signal class.

B. Salinas

The Salinas image shown in Fig. 13 is also an AVIRIS image collected over Salinas Valley, California, and with a spatial resolution of 3.7 m per pixel with a spectral resolution of 10 nm. The image cube has a size of 512 × 217 × 224. This scene is very similar to the Purdue Indiana Indian Pines scene, which also includes 20 water absorption bands, 108–112, 154–167, and 224. Fig. 13(b) and (c) shows the color composite of the Salinas image and the corresponding ground truth map in Fig. 13(b) along with color class labels in Fig. 13(c).

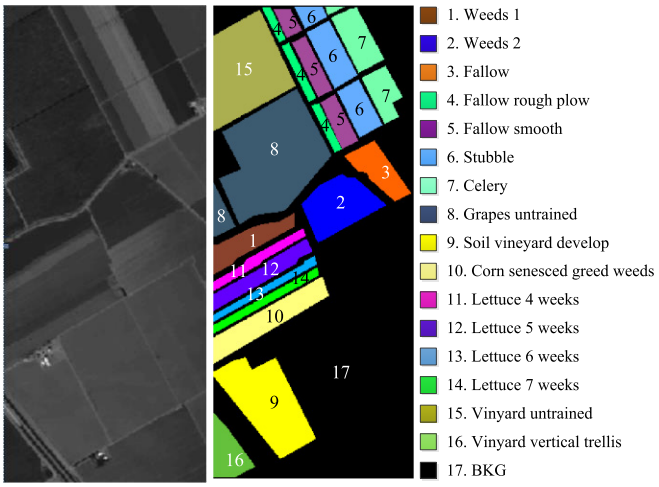
Fig. 14 shows the spectral profiles of the 16 class means in Fig. 13(c). Fig. 15 shows the 3-D ROC curves from the abundance fractional maps obtained by $\delta_\tau^{\text{EPF-G-g}}(\mathbf{r})$, $\delta_\tau^{\text{IEPF-G-g}}(\mathbf{r})$ plus the 3-D ROC curves produced by ICEM and ILCMV at the last iteration before their iterative processes were terminated. Specifically, three 2-D ROC curves of (P_D, P_F), (P_D, τ), and (P_F, τ) corresponding to their 3-D ROC curves are also plotted for 16 classes. The AUC values from three 2-D ROC curves were also calculated and are shown as legends in Fig. 15, where ICEM and ILCMV produced the highest AUC

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TABLE VI

P_A, P_{MC}, P_{AA}, P_{OA}, P_{OPR}, AND $P_{PR(BKG)}$ RESULTING FROM EPF-G-G, IEPF-G-G, ICEM, AND ILCMV FOR PURDUE'S DATA

Class	EPF-G-g			IEPF-G-g			ICEM			ILCMV		
	P_A	P_{MC}	$P_{PR(BKG)}$	P_A	P_{MC}	$P_{PR(BKG)}$	P_A	P_{MC}	$P_{PR(BKG)}$	P_A	P_{MC}	$P_{PR(BKG)}$
1	0.9978±0.00	0.0024±0.00	0.4781±0.06	0.9869±0.01	0.0028±0.00	0.4406±0.07	0.9565	0.0008	0.7333	0.9565	0.0006	0.7719
2	0.8228±0.03	0.0283±0.00	0.6827±0.04	0.9464±0.01	0.0322±0.00	0.6822±0.02	0.9734	0.0045	0.9417	0.9608	0.0061	0.9196
3	0.7716±0.10	0.0138±0.00	0.6965±0.04	0.9763±0.01	0.0145±0.00	0.7356±0.03	0.9807	0.0028	0.9346	0.9699	0.0046	0.8974
4	1±0	0.0194±0.00	0.3811±0.07	1±0	0.0059±0.00	0.6634±0.07	0.9831	0.0004	0.9628	0.9873	0.0009	0.9286
5	0.9606±0.01	0.0550±0.01	0.2963±0.04	0.9815±0.01	0.0571±0.01	0.2947±0.04	0.9441	0.0015	0.9383	0.8944	0.0016	0.9290
6	0.9924±0.00	0.0578±0.01	0.3853±0.03	0.9947±0.00	0.0367±0.00	0.4988±0.05	0.9836	0.0047	0.8831	0.9712	0.0039	0.8997
7	0.9642±0.00	0.0005±0.00	0.7187±0.02	0.9642±0.00	0.0004±0.00	0.7462±0.01	1.0000	0.0007	0.6512	1.0000	0.0004	0.7568
8	1.±0	0.0085±0.00	0.7308±0.01	1±0	0.0070±0.00	0.7674±0.02	0.9895	0.0004	0.9834	0.9874	0.0010	0.9574
9	0.8700±0.12	0.0002±0.00	0.7561±0.10	1±0	0.0011±0.02	0.4511±0.01	1.0000	0.0008	0.5405	1.0000	0.0009	0.5263
10	0.9021±0.06	0.0256±0.00	0.6352±0.06	0.9471±0.02	0.0162±0.00	0.7415±0.05	0.9527	0.0032	0.9344	0.9393	0.0040	0.9185
11	0.8951±0.03	0.0358±0.01	0.7719±0.05	0.9536±0.01	0.0322±0.00	0.7973±0.03	0.9756	0.0059	0.9565	0.9470	0.0078	0.9417
12	0.9463±0.01	0.0242±0.00	0.5380±0.06	0.9925±0.00	0.0228±0.00	0.5653±0.06	0.9663	0.0018	0.9409	0.9562	0.0033	0.8943
13	0.9941±0.00	0.0034±0.00	0.7381±0.02	0.9931±0.00	0.0025±0.00	0.7911±0.02	0.9756	0.0003	0.9662	0.9854	0.0007	0.9352
14	0.9660±0.01	0.1276±0.01	0.3272±0.01	0.9888±0.01	0.1368±0.01	0.3174±0.01	0.9747	0.0015	0.9770	0.9352	0.0019	0.9689
15	0.9795±0.01	0.1665±0.02	0.1003±0.01	1±0	0.1745±0.01	0.0976±0.00	0.9767	0.0007	0.9642	0.9067	0.0006	0.9669
16	1±0	0.0084±0.00	0.3566±0.06	0.9763±0.01	0.0054±0.00	0.4676±0.09	0.9785	0.0007	0.8667	0.9892	0.0007	0.8598
P_{AA}		0.9414±0.01			0.9806±0.00			0.9757			0.9615	
P_{OA}		0.9114±0.00			0.9698±0.00			0.9729			0.9509	
P_{OPR}		0.4443±0.00			0.4727±0.00			0.9426			0.9264	



(a) Salinas scene (b) ground-truth image (c) classes by colors

Fig. 13. Salinas and its ground truth with 16 classes. (a) Salinas scene. (b) Ground truth image. (c) Classes by colors.

values of 2-D ROC curves of (P_D, P_F). Conversely, IEPF-G-g and EPF-G-g produced the highest detection power/probability specified by the AUC values of 2-D ROC curves of (P_D, τ). As for BKG suppression, the best classifier was ICEM with the lowest AUC values of 2-D ROC curves of (P_F, τ) compared to ILCMV with the highest AUC values of 2-D ROC curves of (P_F, τ).

Table VIII tabulates P_A, P_{MC}, and $P_{PR(BKG)}$ produced by EPF-G-g in [23] and IEPF-G-g developed in [49] where the results were calculated by ten runs using randomly selected training samples in the same way done in [23] with $\pm$ standard deviations. Table VIII also includes the results of P_A, P_{MC}, P_{AA}, P_{OA}, P_{OPR}, and $P_{PR(BKG)}$ produced by ICEM and ILCMV for comparison where no $\pm$ standard deviations were calculated because ICEM and ILCMV used class means not randomly selected training samples as EPF and IEPF did. Since both ICEM and ILCMV are mixed pixel classifiers, they used Otsu's method to find appropriate threshold values $\tau_{\text{Otsu}}^{\text{CEM/LCMV}}(\mathbf{r})$ in (43) for classification. According to Table VIII, we could conclude that IEPF-G-g was the best classifier with ILCMV being the worst classifier. Interestingly, EPF-G-g did better in P_{AA} than ICEM but worse in P_{OA} than ICEM. However, this is not the whole story. If we examined P_{OPR}, ICEM and ILCMV produced P_{OPR} of 95% and 93% which were nearly twice much (46% and 47%) produced by EPF-G-g and IEPF-G-g. So, if we factor P_{AA}, P_{OA}, and P_{OPR} in classification performance, ICEM and ILCMV performed very well on the whole and produced more than 93% accuracy of all the three measures with ICEM as the best classifier and EPF-G-g as the worst classifier.

C. University of Pavia

The third hyperspectral image data shown in Fig. 16(a) were collected and recorded by the ROSIS-03 satellite sensor over an urban area surrounding the University of Pavia. It is of size 610 × 340 × 115 with a spatial resolution of 1.3 m per pixel and a spectral coverage ranging from 0.43 to 0.86 μm with a spectral resolution of 4 nm (12 most noisy channels were removed before experiments). Nine classes of interest are considered for this image. Fig. 16(b) provides its ground truth map along with color class labels in Fig. 16(c). Table IX also tabulates the number of data samples in parentheses collected for each class.

Fig. 17 plots spectral signaturs of 9 class means in Fig. 16(c). Fig. 18 shows 3-D ROC curves for all nine classes from the abundance fractional maps obtained by $\delta_\tau^{\text{EPF-G-g}}(\mathbf{r})$, $\delta_\tau^{\text{IEPF-G-g}}(\mathbf{r})$ plus the 3-D ROC curves produced by ICEM and ILCMV at the last iteration before their iterative processes were terminated. Specifically, the AUC values from three 2-D ROC curves were also calculated and are shown as legends in Fig. 15.

Table X tabulates P_A, P_{MC}, and $P_{PR(BKG)}$ produced by EPF-G-g in [23] and IEPF-G-g developed in [49] where the results were calculated by ten runs using randomly selected training samples in the same way done in [23] with $\pm$ standard

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TABLE VII
CLASS LABELS OF CLASSES IN SALINAS

class 1 (2009)	Brocoli_green_weeds_1	class 10 (3278)	Corn_senesced_green_weeds
class 2 (3726)	Brocoli_green_weeds_2	class 11 (1068)	Lettuce_romaine_4wk
class 3 (1976)	Fallow	class 12 (1927)	Lettuce_romaine_5wk
class 4 (1394)	Fallow_rough_plow	class 13 (916)	Lettuce_romaine_6wk
class 5 (2678)	Fallow_smooth	class 14 (1070)	Lettuce_romaine_7wk
class 6 (3959)	Stubble	class 15 (7268)	Vinyard_untrained
class 7 (3579)	Celery	class 16 (1807)	Vinyard_vertical_trellis
class 8 (11271)	Grapes_untrained	class 17 (56975)	BKG
class 9 (6203)	Soil_vinyard_develop		

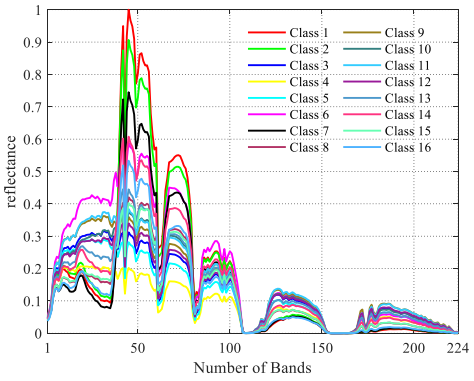


Fig. 14. Spectral profiles of 16-class sample means in Salinas.

deviations. Table X also includes the results of P_A, P_{MC}, P_{AA}, P_{OA}, P_{OPR}, and $P_{PR(BKG)}$ produced by ICEM and ILCMV for comparison where no $\pm$ standard deviations were calculated because ICEM and ILCMV used class means not randomly selected training samples as EPF and IEPF did. Since both ICEM and ILCMV are mixed pixel classifiers, they used Otsu's method to find appropriate threshold values $\tau_{\text{Otsu}}^{\text{CEM/LCMV}}(\mathbf{r})$ in (43) for classification. According to Table X, we could conclude that IEPF-G-g was the best classifier followed by ICEM and ILCMV with the worst results produced by EPF-G-g.

IX. COMPLEXITY ANALYSIS AND DISCUSSION

The 3-D ROC analysis proposed in this article makes use of three parameters, P_D, P_F, and the threshold τ to produce a 3-D ROC curve of (P_D, P_F, τ) from which three 2-D ROC curves of (P_D, P_F), (P_D, τ), and (P_F, τ) can be generated. Interestingly, each of these 2-D ROC curves has its own merit to address various issues in classification.

A. MC Errors

Over the past years, HSIC has been performed in such a way that for the given ground truth, the classification performance is evaluated by P_{OA} or P_{AA} via cross validation with exclusion of BKG from classification. That is, a classifier is validated by ground truth without accounting for BKG in classification. So, if it performs well, does it also imply that it would perform well on the same entire image data set including BKG? The answer to it is obviously not necessarily true as demonstrated in Tables VI, VIII, and X via P_{PR} where EPF-G-g/IEPF-G-g methods had very high P_{OA} values but very poor and low P_{PR} values. To address this issue, we can look into two types of MC error rate, P_{MC} defined in (21) and (22). One error rate, which is referred to as type-I error in the detection theory and also known as false alarm rate, P_F in (9), is caused by misclassifying data samples in the BKG class into the signal class. The other error rate, which is referred to as type-II error and also known as P_M in (8) in the detection theory, is an error resulting from misclassifying data samples in the signal class into the BKG class. So, P_{MC} is the sum of these two types of errors, i.e., $P_{MC} = P_F + P_M$. However, according to (8)

$$P_D + P_M = 1 \Rightarrow P_M = 1 - P_D \tag{45}$$

which implies that P_M is already taken care of by P_D. That is, once P_D is determined, so is P_M. On the other hand, from Fig. 4

$$P_F + P_{PR} = 1 \Rightarrow P_F = 1 - P_{PR} \tag{46}$$

which indicates that P_F can also be determined by P_{PR} in (10) in the same way that P_M is determined by P_D. As a consequence, P_{MC} is in fact specified by P_F. According to (45) and (46), classification performance can also be evaluated by P_D, which determines class accuracy and P_F which specifies P_{MC} as it is used in Tables VI, VIII, and X. Specifically, we can interpret the BKG data samples misclassified into a class of interest as falsely alarmed data samples; the 2-D ROC curve of (P_F, τ) shows the BKG suppression resulting from MC of BKG data samples into a class of interest. As a result, the 2-D ROC curve of (P_F, τ) can be used to address the BKG issue.

B. Class Accuracy

Since P_{OA} is calculated along the diagonal line in the confusion matrix in Fig. 6 which has nothing to do with MC errors, P_{OA} could not effectively address the imbalanced issue. In order to address this issue, Zhong *et al.* [49] introduced a new concept of class features to take care of small classes which would not be sacrificed or compromised by large class. The 2-D ROC curve of (P_D, τ) provides an alternative approach to dealing with the imbalanced class issue by formulating a small class as a desired signal to be detected in noise via a binary hypothesis testing formulation. In this case, the 2-D ROC curve of (P_D, τ) can be used to evaluate the accuracy of individual classes of interest such as small

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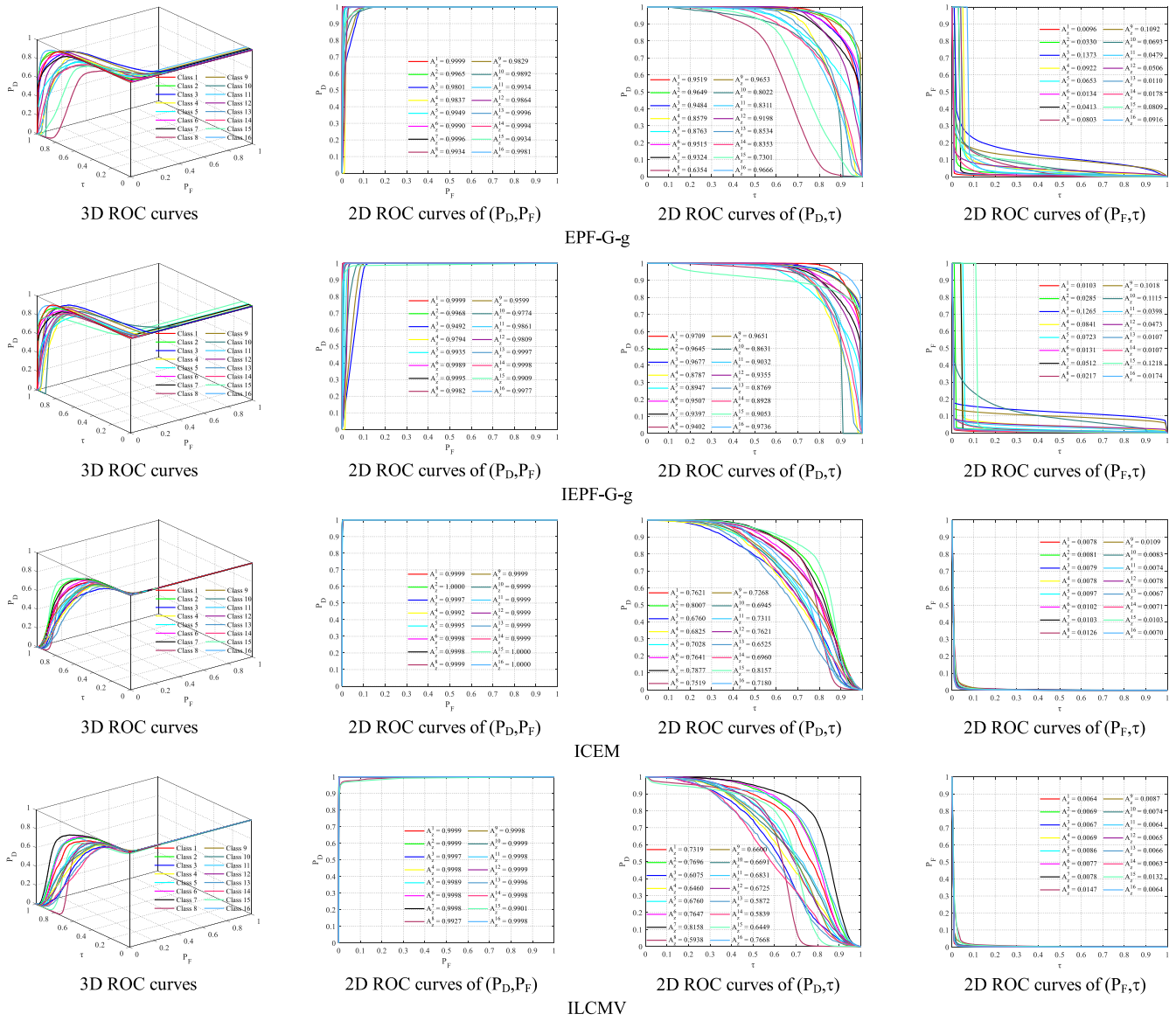


Fig. 15. 3-D ROC analysis produced by EPF-G-g, IEPF-G-g, ICEM, and ILCMV using Otsu's method for Salinas.

classes by $P_A(C_j)$ in (18). That is, the SD probability/power, P_D, can be interpreted as the classification accuracy, $P_A(C_j)$.

With all these being said above, it explains why the 3-D ROC curve can be used to effectively address imbalanced class and BKG issues.

C. Effectiveness of a Soft-Decision Classifier

As shown in Figs. 12, 15, and 18, EPF-G-g, IEPF-G-g, ICEM, and ILCMV performed quite differently. This is mainly due to the FCMA issue resulting from conversion of soft decisions to hard decisions made by the test classifiers. How to evaluate their effectiveness in classification is very challenging. The AUC value of a 2-D ROC curve of (P_D, P_F) offers an objective measure for their performance evaluation. As shown in Fig. 3, two classifiers may generate two different ROC curves, but they indeed have the same effectiveness of detection power/probability by producing the same AUC value. Nevertheless, a 2-D ROC curve of (P_D, P_F) can only be used to evaluate the overall performance of a classifier like P_{OA} used by a confusion matrix. Using it alone cannot describe the entire capability of a classifier in classification. As shown in Figs. 12, 15, and 18, the AUC values of 2-D ROC curves of (P_D, P_F) were very high, but the AUC values calculated for the 2-D ROC curves of (P_D, τ) were very low and poor. This is mainly due to the fact that the threshold τ used to plot the 2-D ROC curve of (P_D, P_F) is actually the same value as used to calculate both P_D and P_F compared to the 2-D ROC curve of (P_D, τ) and 2-D ROC curve of (P_F, τ), which are calculated and plotted by different values of the threshold τ to be considered as an independent variable. This provides evidence that the 3-D ROC analysis is needed to evaluate the performance of a classifier in all aspects.

D. Running Time Complexity

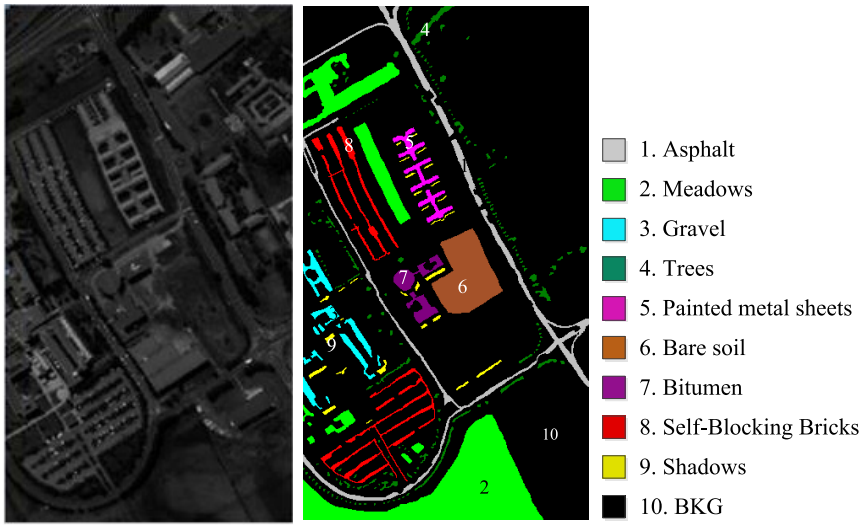
Table XI tabulates the running times in seconds required by EPF-G-g, IEPF-G-g, ICEM, and ILCMV for three

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TABLE VIII

P_A, P_{MC}, P_{AA}, P_{OA}, P_{OPR}, AND $P_{PR(BKG)}$ RESULTING FROM EPF-G-G, IEPF-G-G, ICEM, AND ILCMV FOR SALINAS

Class	EPF-G-g			IEPF-G-g			ICEM			ILCMV		
	P_A	P_{MC}	$P_{PR(BKG)}$	P_A	P_{MC}	$P_{PR(BKG)}$	P_A	P_{MC}	$P_{PR(BKG)}$	P_A	P_{MC}	$P_{PR(BKG)}$
1	1±0	0.0064±0.00	0.7418±0.00	1±0	0.0062±0.00	0.7467±0.00	0.9667	0.0008	0.9562	0.9572	0.0010	0.9473
2	0.9999±8.48	0.0261±0.00	0.5758±0.06	0.9997±0.00	0.0255±0.00	0.5807±0.05	0.9842	0.0006	0.9834	0.9807	0.0008	0.9757
3	0.9981±0.00	0.1396±0.00	0.1147±0.00	0.9998±0.00	0.1344±0.00	0.1189±0.00	0.9074	0.0011	0.9378	0.9408	0.0013	0.9295
4	0.9989±0.00	0.0330±0.00	0.2784±0.01	0.9969±0.00	0.0318±0.00	0.2873±0.03	0.9498	0.0018	0.8688	0.9613	0.0014	0.8993
5	0.9865±0.00	0.0191±0.00	0.5667±0.05	0.9896±0.00	0.0205±0.00	0.5479±0.04	0.9492	0.0026	0.9011	0.9642	0.0033	0.8794
6	0.9999±7.98	0.0088±0.00	0.8077±0.02	1±0	0.0079±0.00	0.8228±0.00	0.9838	0.0029	0.9258	0.9831	0.0028	0.9276
7	0.9997±0.00	0.0081±0.00	0.8039±0.00	0.9992±0.00	0.0077±0.00	0.8109±0.00	0.9869	0.0033	0.9084	0.9799	0.0025	0.9280
8	0.8319±0.04	0.0163±0.00	0.8532±0.04	0.8874±0.03	0.0127±0.00	0.8914±0.05	0.9775	0.0019	0.9835	0.9450	0.0069	0.9390
9	0.9971±0.00	0.1051±0.00	0.3597±0.01	0.9951±0.00	0.0988±0.00	0.3733±0.01	0.9499	0.0004	0.9926	0.9634	0.0011	0.9811
10	0.9643±0.01	0.0833±0.02	0.2689±0.04	0.9892±0.01	0.0974±0.01	0.2406±0.03	0.9652	0.0005	0.9844	0.9539	0.0006	0.9799
11	0.9949±0.00	0.0277±0.00	0.2616±0.02	0.9964±0.00	0.0298±0.00	0.2485±0.03	0.9766	0.0010	0.9030	0.9625	0.0011	0.8916
12	1±0	0.0507±0.00	0.2599±0.02	1±0	0.0437±0.00	0.2901±0.02	0.9704	0.0007	0.9600	0.9699	0.0013	0.9308
13	0.9906±0.00	0.0039±0.00	0.6742±0.00	0.9945±0.00	0.0041±0.00	0.6651±0.01	0.9520	0.0007	0.9237	0.9531	0.0017	0.8259
14	0.9969±0.00	0.0051±0.00	0.6559±0.03	0.9954±0.00	0.0048±0.00	0.6668±0.03	0.9850	0.0008	0.9246	0.9467	0.0014	0.8643
15	0.8787±0.06	0.0194±0.00	0.7636±0.06	0.9595±0.02	0.0146±0.00	0.8233±0.04	0.9796	0.0005	0.9923	0.9250	0.0042	0.9384
16	0.9933±0.00	0.0039±0.00	0.8107±0.06	0.9912±0.00	0.0040±0.00	0.8132±0.08	0.9757	0.0005	0.9681	0.9358	0.0011	0.9337
P_{AA}		0.9769±0.00			0.9871±0.00			0.9662			0.9576	
P_{OA}		0.9449±0.01			0.9687±0.00			0.9697			0.9552	
P_{OPR}		0.4603±0.00			0.4719±0.00			0.9605			0.9373	



(a) University of Pavia scene (b) ground-truth map (c) classes by colors

Fig. 16. University of Pavia and its ground truth with nine classes. (a) University of Pavia scene. (b) Ground truth map. (c) Classes by colors.

TABLE IX

CLASS LABELS OF THE UNIVERSITY OF PAVIA

class 1 (6631)	Asphalt	class 5 (1345)	Painted metal sheets	Class 9 (947)	Shadows
class 2 (18649)	Meadows	class 6 (5029)	Bare Soil	Class 10 (164624)	BKG
class 3 (2099)	Gravel	class 7 (1330)	Bitumen		
class 4 (3064)	Trees	class 8 (3682)	Self-Blocking Bricks		

hyperspectral real image scenes, the Purdue data, Salinas, and University of Pavia used for experiments where the times were calculated by ten runs using randomly selected training samples in the same way done in [23] with ± standard deviations. All experiments were implemented using MATLAB with version 2015b with the computer environment: Intel i7-6700 3.4-Hz-frequency CPU and 16-GB 2133-MHz memory.

As we can see from Table XI for Purdue's data, IEPF-G-g required running time five times that required by EPF-G-g because of the iterative processes implemented in IEPF-G-g. Interestingly, ICEM and ILCMV only required 1/3 and 1/18 of that required by EPF-G-g, respectively, even though they both used iterative processes. This was because ILCMV classified all classes at once while ICEM classified one class at a time. Moreover, ICEM and ILCMV only required calculating class means without involving a process of randomly selecting training samples which also required more time. In addition, since IEPF-G-g required iteratively implementing EPF-G-g, it ended up the most time-consuming classifier.

Also, from Table XI for Salinas, the running time required by IEPF-G-g was almost three times that required by EPF-G-g, ICEM, and ILCMV because it required implementing

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TABLE X

P_A, P_{MC}, P_{AA}, P_{OA}, P_{OPR}, AND $P_{PR(BKG)}$ RESULTING FROM EPF-G-G, IEPF-G-G, ICEM, AND ILCMV FOR THE UNIVERSITY OF PAVIA

Class	EPF-G-g			IEPF-G-g			ICEM			ILCMV		
	P_A	P_{MC}	$P_{PR(BKG)}$	P_A	P_{MC}	$P_{PR(BKG)}$	P_A	P_{MC}	$P_{PR(BKG)}$	P_A	P_{MC}	$P_{PR(BKG)}$
1	0.9803±0.00	0.1465±0.00	0.1812±0.00	0.9943±0.00	0.1472±0.00	0.1828±0.00	0.7702	0.0283	0.4730	0.7667	0.0518	0.3285
2	0.9896±0.00	0.1592±0.01	0.3813±0.01	0.9968±0.00	0.1520±0.01	0.3948±0.02	0.8581	0.0089	0.9054	0.8190	0.0214	0.7909
3	0.9497±0.02	0.0183±0.00	0.3476±0.02	0.9949±0.00	0.0226±0.00	0.3140±0.03	0.9085	0.0063	0.5952	0.6027	0.0087	0.4134
4	0.9516±0.00	0.1319±0.00	0.0978±0.00	0.9661±0.00	0.1783±0.01	0.0756±0.00	0.6586	0.0201	0.3300	0.7882	0.0318	0.2711
5	0.9999±0.00	0.0112±0.00	0.3780±0.07	0.9998±0.00	0.0116±0.00	0.3664±0.05	0.9978	0.0034	0.6556	0.9859	0.0030	0.6793
6	0.9999±0.00	0.2217±0.01	0.1010±0.00	1±0	0.1632±0.01	0.1330±0.01	0.9328	0.0004	0.9841	0.7149	0.0058	0.7527
7	0.9984±0.00	0.0114±0.00	0.3667±0.05	1±0	0.0126±0.00	0.3461±0.05	0.9669	0.0016	0.7983	0.8594	0.0017	0.7625
8	0.9757±0.01	0.0698±0.00	0.2020±0.00	0.9932±0.00	0.0859±0.01	0.1750±0.02	0.8558	0.0174	0.4699	0.7947	0.0331	0.3023
9	0.9997±0.00	0.0548±0.00	0.0776±0.00	0.9995±0.00	0.0482±0.00	0.0880±0.01	0.8543	0.0069	0.3621	0.7835	0.0068	0.3454
P_{AA}		0.9828±0.00			0.9938±0.00			0.8670			0.7739	
P_{OA}		0.9843±0.00			0.9944±0.00			0.8489			0.7764	
P_{OPR}		0.2030±0.00			0.2051±0.00			0.6584			0.5054	

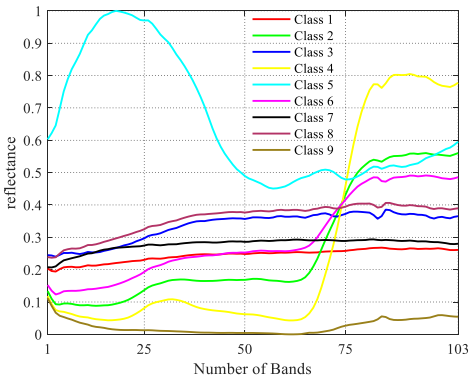


Fig. 17. Spectral profiles of nine-class sample means in the University of Pavia.

TABLE XI

RUNNING TIMES REQUIIRED BY EPF-G-G, IEPF-G-G, ICEM, AND ILCMV FOR PURDUE DATA, SALINAS, AND UNIVERSITY OF PAVIA

	EPF-G-g	IEPF-G-g	ICEM	ILCMV
Purdue	141.6±5.9	715.3±100.5	46.4	7.9
Salinas	163.2±4.4	498.4±15.0	133.7	123.9
U. Pavia	191.1±11.6	643.1±37.2	233.7	64.8

EPF-G-g iteratively. Similarly, for the University of Pavia in Table XI, IEPF-G-g also required the worst running time compared to ILCMV which required the least amount of time with only 1/10 of its running time. EPF-G-g and ICEM were right in between with 3–4 times that required by ILCMV.

X. UTILITY OF ROC ANALYSIS IN CLASSIFICATION

Although the ROC results in Section VII can be used to evaluate the four test classifiers, EPF-G-g, IEPF-G-g, ICEM, and ILCMV, for quantitative studies, the results vary class by class in a wide range. It is difficult to assess the overall classification performance. So, with all things considered in Sections VIII-A–VIII-C, a general overall classification performance of a classifier should include its ROC results and results obtained by classification confusion matrix in Fig. 6 such as P_{OA}, P_{AA}, and P_{OPR}.

More specifically, we can consider the AUC values of the 2-D ROC curves of (P_D, P_F), (P_D, τ), and (P_F, τ) as soft classification measures, while P_{AA}, P_{OA}, and P_{OPR} are considered as hard classification measures. If we further average the three soft classification measures, AUC values of the 2-D ROC curves of (P_D, P_F), (P_D, τ), and (P_F, τ) in Figs. 12, 15, and 18 over all classes and their calculated AUC class means combined with the three hard classification measures, P_{AA}, P_{OA}, and P_{OPR} in Tables VI, VIII, and X allow us to calculate a score to measure the overall performance of a classifier by

$$\text{score} = \text{AUC}(P_D, P_F) + \text{AUC}(P_D, \tau) - \text{AUC}(P_F, \tau) + P_{AA} + P_{OA} + P_{OPR} \quad (47)$$

where the values of $\text{AUC}(P_D, P_F)$, $\text{AUC}(P_D, \tau)$, P_{AA}, P_{OA}, and P_{OPR} are summed with subtraction of $\text{AUC}(P_F, \tau)$. This is because a higher value of $\text{AUC}(P_D, P_F)$, $\text{AUC}(P_D, \tau)$, P_{AA}, P_{OA}, and P_{OPR} indicates a higher classification performance. Conversely, a lower value of $\text{AUC}(P_F, \tau)$ indicates a better BKG suppression and, thus, a better classification performance. As a result, (47) indeed combines the values calculated by three soft measures and three hard measures, respectively, to produce a quantitative value of the overall classification performance for each of four classifiers as a score for comparison. Tables XII–XIV tabulate the scores calculated by (47) for the four test classifiers, EPF-G-g, IEPF-G-g, ICEM, and ILCMV for the three image data sets, Purdue's data, Salinas, and University of Pavia, respectively.

According to the results in Tables XII–XIV, we can draw the following conclusions.

1) On the one end, if we only use the three soft classification measures provided by AUC values of the 2-D ROC curves of (P_D, P_F), (P_D, τ), and (P_F, τ), the mixed pixel classifiers, ICEM and ILCMV, performed significantly better than pure-pixel classifiers, EPF-G-g and IEPF-G-g. This is particularly true for Purdue's data, which has the imbalance class issue. This indicates that the soft classification measures can be more effective to address the imbalanced class, FCMA and BKG issues, in terms of P_{OPR}, the 2-D ROC curves of (P_D, P_F), and the 2-D ROC curve of (P_F, τ), respectively.

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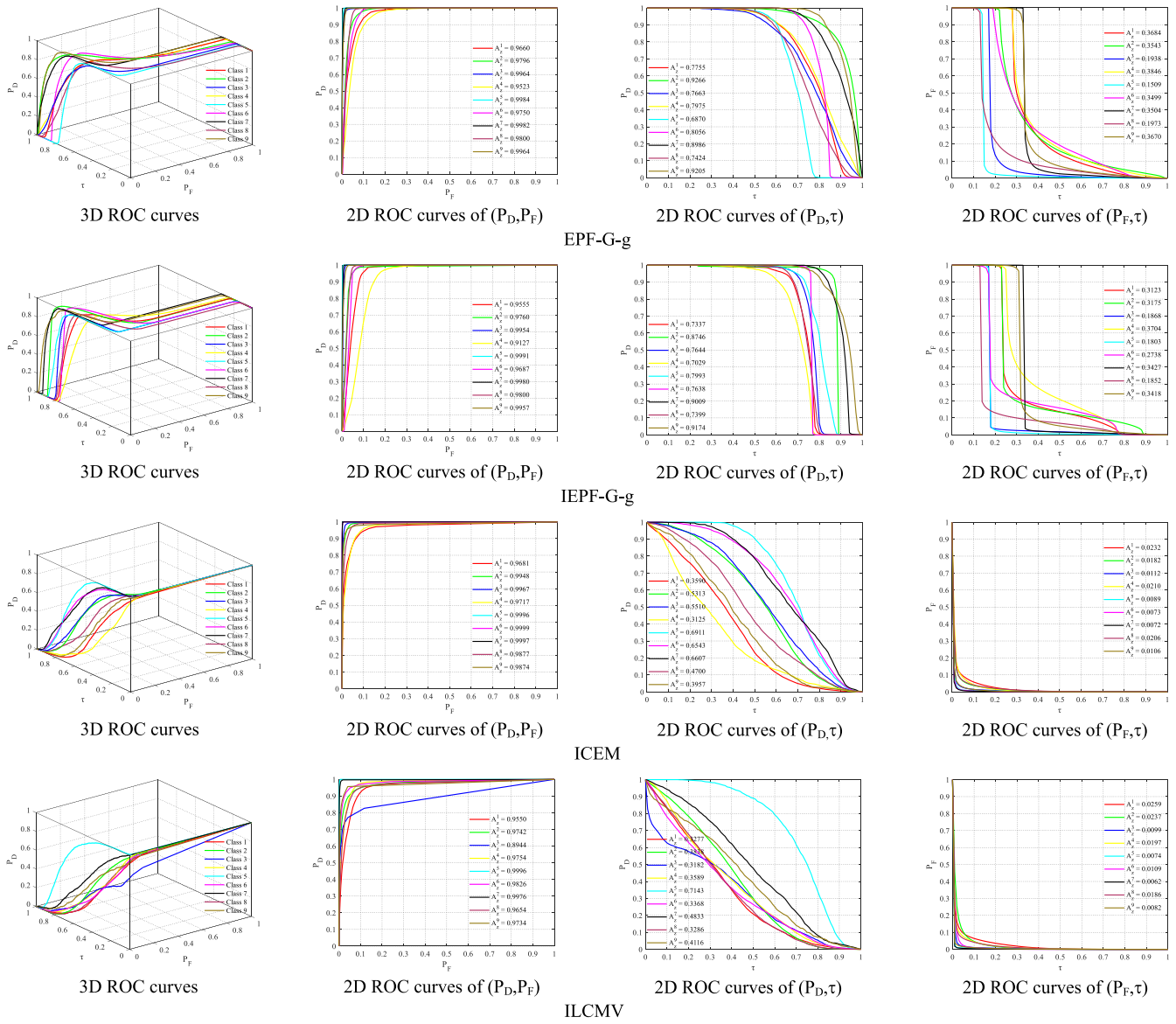


Fig. 18. 3-D ROC analysis produced by EPF-G-g, IEPF-G-g, ICEM, and ILCMV using Otsu's method for the University of Pavia.

2) On the other end, if hard classification measures, P_{OA} and P_{AA}, are used, IEPF-G-g outperformed EPF-G-g, ICEM, and ILCMV, whereas EPF-G-g also performed reasonably well. This shows that the hard classification measures are good for pure pixel-based classifiers. This fact also explains why P_{OA} and P_{AA} have been used for HSIC because the used spectral classifiers are generally pure pixel-based such as SVM. However, it is interesting to see that ICEM and ILCMV produced the values of P_{OPR} 2–3 times better than that produced by EPF-G-g and IEPF-G-g. This is because P_{OPR} is used to evaluate the effectiveness of a classifier based on its classified data samples which includes BKG samples. The details on discussions of this issue can be found in [28], [29], and [46].
3) If both soft and hard classification measures are combined to evaluate a classifier, it clearly showed that the best classifier was ICEM across board for all the three image data sets. For the two AVIRIS data sets, Purdue's data and Salinas, ICEM and ILCMV outperformed EPF-G-g and IEPF-G-g with EPF-G-g being the worst classifier. Surprisingly, for the University of Pavia, EPF-G-g, which was the worst classifier for Purdue's data and Salinas, performed better than ILCMV and only slightly worse than IEPF-G-g. This is mainly due to the fact that the University of Pavia has most complicated BKG and EPF-G-g produced its highest AUC value of its 2-D ROC curves of (P_{D}, τ) at the expense of highest AUC value of its 2-D ROC curves of (P_{F}, τ). In this case, ILCMV produced the worst AUC values because of the FCMA issue. Nevertheless, ICEM still performed well and better than IEPF-G-g and EPF-G-g.

As a concluding remark, how a classifier performs in classification should be evaluated by both soft and hard measures. The results solely relying on P_{OA} and P_{AA} may be misleading.

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TABLE XII
SCORES CALCULATED BY (47) FOR EPF-G-G, IEPF-G-G, ICEM, AND ILCMV IN TERMS OF ROC ANALYSIS, P_{AA}, P_{OA}, AND P_{OPR} FOR PURDUE'S DATA

	EPF-G-g	IEPF-G-g	ICEM	ILCMV
AUC(P_D,P_F)	0.9928	0.9967	0.9998	0.9995
AUC(P_D,τ)	0.7565	0.8108	0.7373	0.6631
AUC(P_F,τ)	0.1732	0.1426	0.0140	0.0105
P_{AA}	0.9414	0.9806	0.9757	0.9615
P_{OA}	0.9114	0.9698	0.9729	0.9509
P_{OPR}	0.4443	0.4727	0.9426	0.9264
Score	3.8732	4.0879	4.6143	4.4909

TABLE XIII
SCORES CALCULATED BY (47) FOR EPF-G-G, IEPF-G-G, ICEM, AND ILCMV IN TERMS OF ROC ANALYSIS, P_{AA}, P_{OA}, AND P_{OPR} FOR SALINAS

	EPF-G-g	IEPF-G-g	ICEM	ILCMV
AUC(P_D,P_F)	0.9931	0.9880	0.9998	0.9987
AUC(P_D,τ)	0.8764	0.9264	0.7328	0.6796
AUC(P_F,τ)	0.0594	0.0543	0.0087	0.0080
P_{AA}	0.9769	0.9871	0.9662	0.9576
P_{OA}	0.9449	0.9687	0.9697	0.9552
P_{OPR}	0.4603	0.4603	0.9605	0.9373
Score	4.1922	4.2762	4.6203	4.5204

TABLE XIV
SCORES CALCULATED BY (47) FOR EPF-G-G, IEPF-G-G, ICEM, AND ILCMV IN TERMS OF ROC ANALYSIS, P_{AA}, P_{OA}, AND P_{OPR} FOR UNIVERSITY OF PAVIA

	EPF-G-g	IEPF-G-g	ICEM	ILCMV
AUC(P_D,P_F)	0.9825	0.9757	0.9895	0.9686
AUC(P_D,τ)	0.8133	0.7997	0.5140	0.4070
AUC(P_F,τ)	0.3019	0.2790	0.0142	0.0145
P_{AA}	0.9828	0.9938	0.8670	0.7739
P_{OA}	0.9843	0.9944	0.8489	0.7764
P_{OPR}	0.2030	0.2051	0.6584	0.5054
Score	3.6640	3.6897	3.8635	3.4168

XI. CONCLUSION

This article presents a new evaluation tool for HSIC, called 3-D ROC analysis, which can effectively address the issues caused by BKG, imbalanced classes, and FCMA through its generated three 2-D ROC curves. By formulating a classification problem as a binary composite hypothesis testing problem, a class to be classified and all other classes plus BKG can be specified by the alterative hypothesis as a signal to be detected and null hypothesis as noise, respectively. In this case, the 2-D ROC curve of (P_F, τ) calculated from a 3-D ROC curve can be used to evaluate the impact of the BKG suppression on classification. On the other hand, using the same hypothesis testing formulation, we can interpret the power/probability of SD as accuracy of a class to be classified regardless of whether the class is small or large, while all other classes plus BKG as noise. Consequently, the 2-D ROC curve of (P_D, τ) calculated from a 3-D ROC curve can be used to address the imbalance class issue. Finally, converting soft decisions made on FCMA by a classifier to hard decisions by varying a threshold τ extends the traditional 2-D ROC curve of (P_D, P_F) calculated to a 3-D ROC curve that can be applicable to assessing effectiveness of a classifier on the FCMA issue. Accordingly, the 3-D ROC analysis provides an invaluable tool to measure the performance of a classifier from various aspects. In particular, the performance of HSIC can be evaluated by the 3-D ROC analysis along with its three 2-D ROC curves considered as soft classification measures in conjunction with the commonly used criteria such as P_{OA}, P_{AA} viewed as hard classification measures to further analyze a classifier in various aspects as demonstrated in Tables XII–XIV. So, the development of 3-D ROC analysis using its generated three 2-D ROC analysis to address the issues of BKG, imbalanced class, and FCMA is novelties and major contributions of this article. Last but not least, this article provides 3-D ROC analysis-derived soft classification measures to supplement the statistical detection theory to HSIC developed in [46].

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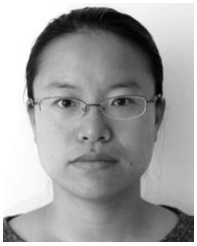
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